

WDW-8

Covariant–Canonical Reconciliation and Controlled Semiclassical Recovery

A source-attached constraint-scaled recovery germ on the compact hyperbolic subtype

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constraintsystems.institute.org | August 2026

Exact scope. From the WDW-6 renormalized Branch-A operator and WDW-7 profile-clock endpoint, the theorem constructs a populated, source-attached semiclassical recovery germ with a common operator domain and explicit error register. The physical theory remains the isolated $\delta = 1$ anchor. All conclusions are local to the declared Einstein–Klein–Gordon subtype and terminate at the printed failure boundary.

Abstract

The recovery of classical spacetime and relational quantum dynamics from the Wheeler–DeWitt constraint requires more than a formal WKB ansatz: it requires a populated comparison family, common operator domains, a noncharacteristic phase chart, a canonical-to-covariant reconstruction, quotient-compatible comparison maps, specified error topologies, and a failure boundary. These requirements are established here on the source-generated compact hyperbolic homogeneous Einstein–Klein–Gordon subtype inherited from WDW-1–7.

The physical WDW-6 theory remains an isolated anchor. A distinct comparison germ scales only the quantum part of its renormalized Hamiltonian,

$$\widehat{\mathcal{H}}_\delta[N] = M_V[N] + \delta^2(\widehat{\mathcal{H}}^{\text{ren}}[N] - M_V[N]), \quad \varkappa_\delta = \delta\hbar,$$

on the same graph source and common domain, without changing the clock, source, couplings, boundary class, or classical target. Positive scalar momentum makes the reduced Hamilton–Jacobi problem noncharacteristic. The method of characteristics and the transport equation then produce smooth, cutoff-supported non-anchor states $A \exp(iS/\varkappa_\delta)$ on a compact recovery tube. Exact substitution gives a uniform $O(\delta^2)$ Hamilton–Jacobi remainder. The canonical characteristic through the inherited Cauchy datum independently reconstructs the same source-attached Einstein–Klein–Gordon development, so the Einstein/source, Levi–Civita, and curvature comparison errors vanish exactly.

The conditional carrier is the actual WDW-7 value/normal-response trace $Y_{\delta,\tau}^{\text{rec}} = J_\tau \Psi_\delta$, not a point evaluation or an ungenerated Born–Oppenheimer factor. Reapplication of the scaled constraint gives an exact defect-forced companion equation and hence an induced no-spectator connection with zero conditional-equation residual. The represented scalar algebra and complete volume record agree exactly. For backreaction, a fixed-clock full-envelope metric response is attached before a separate reduced scalar action is varied. A homogeneous constraint right inverse and the history-dependent source $\text{Src}_\varphi^{\text{hom}}(\gamma^\epsilon) \boxplus \epsilon C_\delta^Q[\gamma^\epsilon]$ give an exact constraint-preserving family. The canonical, covariant, and complete source responses are $O(\delta^2)$, the nonlinear remainder is $O(\delta^4)$, and the constraint and total Ward residuals vanish.

The continuation required by the Module-C interface is obtained by advancing the anchored Cauchy embedding with the WDW-4A spacetime scalar flow and mapping it through WDW-3’s shifted-tower evaluation equivalence. This yields the occupied characteristic-plaque line and a locally Lipschitz clock seed without integrating the derivative-losing ADM field; differentiated evaluation only afterward gives $d\pi_{\text{pr}} X_{(\sqrt{\hbar}/p_\phi, 0)}$. The terminal zero exchange ledger is fixed upstream, the metric, quantum-line, record, and compatibility processes factor through the complete state, and their typed closure conditions give history-indexed identity cells and strict functorial four-cubes. Fixed-domain AASC exhaustion then leaves one lawful same-scope recovery class, up to the stated reparameterization and redescription equivalences. The theorem is local and subtype-indexed; it makes no global superspace, global-clock, ultraviolet, singular, positive-frequency, Born–Oppenheimer, or Standard-Model claim.

Scope and logical structure

The paper separates four statements which are frequently conflated:

1. the accepted physical WDW-6 state and WDW-7 conditional endpoint at $\delta = 1$;
2. a populated comparison germ for $0 < \delta \leq \delta_0$;
3. classical canonical data extracted from the leading phase of that germ;
4. comparison of the independently reconstructed covariant history with the accepted WDW-4 history.

The asymptotic states are not claimed to converge in norm to the physical WDW-6 state. Their attachment is instead the one specified by the WDW-8 comparison class: common source ancestry, common graph operator lineage, the physical anchor at $\delta = 1$, value-one normalization at $q_\star$, equality of the Cauchy phase jet, and a fixed classical principal symbol. The parameter space is the disjoint union

$$\mathbf{P}_8 = \{1\} \sqcup (0, \delta_0],$$

so no unproved interpolation between the physical theory and the asymptotic germ is assumed.

Scope of the theorem. The result is a singleton lawful recovery on the fixed Branch-A, compact hyperbolic, homogeneous-cylinder, single-rate, source-generated no-spectator scope. Openly changed sources, jets, carriers, operator grammars, probes, or targets remain standing-bearing enrichments.

Principal theorem

Theorem 0.1 (WDW-8 covariant–canonical reconciliation). *Let $\Sigma = \mathbb{H}^3/\Gamma$ carry the sectional-curvature -1 metric $h_\star$. Fix the accepted WDW-4A datum*

$$z_\star = (h_\star, \pi_\star; 0, \epsilon_\star \sqrt{h_\star}), \quad \epsilon_\star > 0,$$

its source-attached Einstein–Klein–Gordon development $[\gamma_\star]_{\text{src}}$, the accepted WDW-6 Branch-A graph source $\mathcal{V}_A^{\text{ren}}$, normalized joint solution $F_A^{(0)}$, physical line and scalar algebra, and the accepted WDW-7 profile-clock package. On the compact recovery tube and single-rate comparison class defined below:

1. *the family $\widehat{\mathcal{H}}_\delta$ is an actual source-attached common-domain operator family, contains non-anchor members, and satisfies $\widehat{\mathcal{H}}_1 = \widehat{\mathcal{H}}^{\text{ren}}$;*
2. *canonical minimal prolongations generate the initial phase and amplitude data, and there are a real Hamilton principal function S , a positive transport amplitude A , and a fixed cutoff $\chi \equiv 1$ on the comparison tube such that $\Psi_\delta = E_{\text{hom}}(\chi A e^{iS/(\delta h)}) \in \mathcal{V}_A^{\text{ren}}$ is populated for every $0 < \delta \leq \delta_0$;*
3. *for constants independent of δ ,*

$$E_{\text{HJ}}(\delta) \leq C_{\text{HJ}}\delta^2, \quad E_{\text{Ein}}(\delta) = 0, \quad E_{\text{mat}}(\delta) = 0, \quad E_{\text{obs}}(\delta) = 0, \quad E_{\text{back}}(\delta) \leq C_{\text{back}}\delta^2;$$

4. *the canonical characteristic selected by $dS(q_\star)$ reconstructs $[\gamma_\star]_{\text{src}}$ independently of the comparator, and the recovered derivative and curvature are those of the fixed WDW-1 Levi–Civita branch;*
5. *the WDW-7 source-generated line retains $H_{\text{rel}}^{\text{C}} = 0$; the separate recovery line is generated by $J_\tau \Psi_\delta$, descends from the same physical quotient, carries the Hermitian connection forced by the exact inhomogeneous scaled companion identity and a separately typed zero endomorphism, and has exact conditional-equation residual; its represented algebra is $\mathbb{C}I$, and its complete volume record agrees with the reconstructed relational record;*
6. *a homogeneous right inverse, an explicit exact initial constraint curve, a fixed-clock full-envelope Module-A metric response admitted by graph attachment, a separate reduced scalar action, an exact history-dependent total source, and a direct homogeneous Einstein–source theorem construct a constraint-preserving family whose complete canonical, covariant, and source response is $O(\delta^2)$ and whose nonlinear remainder is $O(\delta^4)$; the WDW-4A spacetime clock flow, pushed through WDW-3 evaluated-plaque equivalence, constructs an RPI-level seed and nonidentity Module-C Moore category before target evaluation; the ADM characteristic formula is derived afterward, its anchor flow is proved equal to the actual WDW-7 occupied lineage, and the target-specific metric, quantum-line, record, and compatibility processes factor through the complete upstream state; the zero exchange state ledger and history incidence are independently declared upstream; their recovery maps are natural, the four typed physical closure tests establish occupation, and history-indexed identity cells generate functorial strict closed cubes;*
7. *every fixed-domain candidate is classified at its first decisive condition, typed as an open enrichment, or equivalent to this package. Consequently*

$$\left| \text{Real}_8^{\text{law}} / \sim_8 \right| = 1, \quad G_8 = \textbf{PASS}, \quad \text{Authorize}(\text{WDW-9}) = \text{TRUE}.$$

All conclusions stop at the first loss of the phase chart, scalar-clock transversality, homogeneous-cylinder attachment, common domain, history-dependent source map, local reconstruction chart, characteristic

seed, RPI role occupation, actual-lineage identification, target-process profile, incidence cocycle, exchange cube, compact uniformity tube, or stated remainder bound.

The proof is the ordered chain in Sections 1–7. No item of the principal theorem is used to populate an earlier item.

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1. Fixed target, inherited data, and comparison class

1.1 Dependency order and inheritance boundary

The fixed dependency order is

$$G_0 \longrightarrow G_\nabla \longrightarrow G_1 \longrightarrow G_2 \longrightarrow G_3 \longrightarrow G_4 \longrightarrow G_5 \longrightarrow G_6 \longrightarrow G_7 \longrightarrow G_8.$$

WDW-1 supplies the canonical projection and the source-fibre-preserving local reconstruction theorem. WDW-2 supplies proper spatial redescription descent. WDW-3 identifies the full regular characteristic distribution with the lapse–shift constraint directions and proves finite refoliation coherence. WDW-4A supplies a nonvacuum Einstein–Klein–Gordon occupant and the function-valued clock gauge $\phi(\cdot) = \tau \mathbf{1}$. WDW-5 supplies the configuration-space differential operators. WDW-6 supplies the renormalized graph source, common domain, complete joint solution, positive rank-one physical quotient, represented algebra, and guarded recovery handoff. WDW-7 supplies the source-generated Cauchy-germ line, its constraint-induced connection, the zero covariant line Hamiltonian, and the nonconstant complete volume record.

No WDW-8 construction changes an ordering, counterterm, anomaly class, physical form, clock, boundary class, or source assignment upstream. Failure of the recovery germ leaves the exact WDW-6/7 object intact.

Theorem 1.1 (Inherited data and fixed scope). *The only authorized WDW-8 entry object is*

$$I_{6 \rightarrow 8} = (H_{4,A}^{\text{cov}}(\epsilon_\star), S_6, O_{\text{phys}}, A_{\text{seed}}^{(0)}, C_{\text{src}}^{(0)}, E_{\text{guard}}^{(0)}),$$

where $A_{\text{seed}}^{(0)}$ contains the physical state $F_A^{(0)}$, the covariant source history $[\gamma_\star]_{\text{src}}$ and the authenticated classical Cauchy phase jet, while

$$E_{\text{guard}}^{(0)} = (E_{\text{HJ}}, E_{\text{Ein}}, E_{\text{mat}}, E_{\text{obs}}, E_{\text{back}}; \sigma_{\text{est}} = \textbf{UNESTIMATED-WDW8}).$$

The five displayed E entries are analytic errors; σ_{est} is a status coordinate. The import fixes the compact boundaryless metric–scalar target, proper quotient, profile clock, source, connection branch, physical anchor, and exact failure boundaries. It supplies no recovery family or estimate.

Proof. This is the guarded recovery handoff compiled into the accepted WDW-6 capstone, together with the accepted WDW-7 endpoint. The handoff expressly marks each error unestimated. WDW-7 expressly leaves WKB, Born–Oppenheimer, Einstein, and low-energy recovery to WDW-8. The theorem merely imports those typed coordinates and therefore makes no downstream inference. $\square$

1.2 Fixed geometric and source subtype

Let $h_\star$ be the curvature -1 metric on the closed oriented hyperbolic carrier $\Sigma = \mathbb{H}^3/\Gamma$ and let

$$V_\star = \text{Vol}_{h_\star}(\Sigma), \quad c_\star = \sqrt{1 + \kappa \epsilon_\star^2/6}.$$

The WDW-4A datum is

$$h = h_\star, \quad K_{ij} = c_\star h_{\star ij}, \quad \phi = 0, \quad p_\phi = \epsilon_\star \sqrt{h_\star}.$$

It satisfies the Hamiltonian and momentum constraints exactly and has scalar energy density $\epsilon_\star^2/2$. The scalar normal speed is positive, so the level-set gauge $\phi(\cdot) = \tau \mathbf{1}$ is transverse on a nonempty interval $I_\star$.

Define the homogeneous configuration embedding

$$\iota_{\text{hom}} : (a, \varphi) \longmapsto (a^2 h_\star, \varphi \mathbf{1}), \quad (a, \varphi) \in U_{\text{hom}} \subset (0, \infty) \times \mathbb{R},$$

with $(1, 0) \in U_{\text{hom}}$. This is a recovery subtype, not a claim that the complete WDW source has only two degrees of freedom.

Lemma 1.2 (Occupied hyperbolic homogeneous subtype). *The source-attached development through $z_\star$ preserves the locally homogeneous hyperbolic scalar ansatz on a nonempty clock interval. Its lift to $\mathbb{H}^3$ is an open-FLRW Einstein–Klein–Gordon development, and quotienting by Γ returns the accepted compact history. The characteristic through $(a, \varphi) = (1, 0)$ has*

$$\dot{a}/a = c_\star, \quad \dot{\varphi} = \epsilon_\star$$

in unit lapse at the distinguished leaf.

Proof. Lift the WDW-4A initial data to $\mathbb{H}^3$. They are invariant under the orientation-preserving isometry group of the constant-curvature metric: K is proportional to h , the scalar is spatially constant, and its normal momentum density is the metric volume density times a constant. Acting by an isometry therefore produces the same Cauchy data. Geometric uniqueness of the Einstein–Klein–Gordon Cauchy problem extends these local symmetries to the development, giving the locally FLRW form. The lattice action is included and the quotient returns the compact source-attached development. The two displayed velocities are the defining WDW-4A data. $\square$

1.3 Fixed-domain structure and generated prolongations

The WDW-8 comparison universe is governed by the established AASC fixed-domain results, independently of any preference for a particular WKB seed. The kernel theorem fixes Admissibility, Standing, Reference, and Irreversibility for every nondegenerate determinate package. The Fixed-Domain Exhaustion Theorem (Definition 0.1 and Theorems 7.1–7.4), the same-scope operator exhaustion theorem (Theorem 7.1), and the Bivalence Theorem then give the following specialization [8, 9, 10, 11].

Theorem 1.3 (WDW-8 fixed-domain specialization). *Fix the literal handoff of Theorem 1.1, the compact hyperbolic target, the one attached source C_{src} , the WDW-2/3 quotient, the WDW-4A profile clock, the WDW-5/6 graph source and operator grammar, the WDW-7 physical quotient and complete record, the WDW-1 connection branch, and the five-error use profile. Then every purported same-scope recovery distinction is exactly one of:*

1. *a lawful redescription which factors through the fixed quotient and is bookkeeping-equivalent;*
2. *failure of an already fixed admissibility condition;*
3. *an imported operator, datum, classifier, or auxiliary construction, which is inadmissible at the same scope; or*
4. *a change of carrier, source, clock, target, operator grammar, domain, comparison probe, failure locus, or continuation rule, which is a standing-bearing scope enrichment.*

There is no fifth same-scope admissibility-bearing locus. In particular, an unregistered higher phase jet, amplitude profile, polarization, matter factor, or Hamiltonian selector cannot acquire same-scope standing merely by being written as a function of inherited data.

Proof. The listed WDW objects instantiate the fixed-domain comparison coordinates of Definition 0.1 of the imported exhaustion theorem. Its Theorem 7.3 reduces every admissibility-bearing variation to boundary typing, evaluated act, quotient class, or continuation; Theorem 7.4 removes path- and representative-sensitive residual distinctions. The same-scope operator exhaustion theorem classifies a new operator as bookkeeping-equivalent or inadmissible unless its carrier or operator grammar is changed openly. The Bivalence Theorem excludes an intermediate standing status and the kernel excludes same-act modification. Specializing those exhaustive alternatives to the specified WDW coordinates gives the four rows above. The final sentence is the auxiliary-datum dichotomy applied to data not present in the handoff or the registered local constructor grammar. $\square$

Let $L_0 = \{\varphi = 0\}$ be the noncharacteristic initial line in the source-fixed volume coordinate a . On its connected coordinate interval define

$$j_\star^1 f = (f(1), \partial_a f(1)), \quad e_\star g = g(1),$$

and the two no-spectator subspaces

$$\mathcal{M}_{\text{ph}} := \ker(\partial_a^2), \quad \mathcal{M}_{\text{amp}} := \ker(\partial_a).$$

The derivative operators here are not freely chosen coordinates: a is the dimensionless total-volume coordinate fixed by the occupied homogeneous embedding. Vanishing of $\partial_a^2 f$ says that no phase Hessian or higher initial jet has been added; vanishing of $\partial_a g$ says that no amplitude profile has been added.

Define the prolongation maps

$$P_{\text{ph}}(s, p)(a) := s + p(a - 1), \quad P_{\text{amp}}(c)(a) := c. \quad (1)$$

Theorem 1.4 (Unique generated-minimal prolongations). *P_{ph} is the unique right inverse of $j_\star^1$ with image in $\mathcal{M}_{\text{ph}}$, and P_{amp} is the unique right inverse of $e_\star$ with image in $\mathcal{M}_{\text{amp}}$. In particular, the inherited phase jet $(0, p_{a\star})$ and unit coefficient generate, rather than presuppose,*

$$S_0 = P_{\text{ph}}(0, p_{a\star}), \quad A_0 = P_{\text{amp}}(1).$$

Any different smooth initial function with the same anchor value and first phase jet has nonzero second derivative somewhere, and any different amplitude with the same anchor value has nonzero first derivative somewhere; each therefore carries an additional standing-bearing initial coordinate.

Proof. Every element of $\ker(\partial_a^2)$ is affine on a connected interval, so its value and first derivative at $a = 1$ fix it uniquely as $s + p(a - 1)$. Every element of $\ker(\partial_a)$ is constant and is fixed by its value at $a = 1$. The displayed maps attain those values and are therefore the unique right inverses with the stated images. The final assertion is the contrapositive of these two kernel characterizations. $\square$

1.4 The fixed lawful candidate universe and equivalence

Definition 1.5 (Fixed WDW-8 candidate and lawful residue). Let $\text{Cand}_8^{\text{fix}}$ be the class of attempted recovery packages formed after Theorem 1.3 from the following fixed coordinates:

1. the literal handoff of Theorem 1.1;
2. the parameter carrier $P_8 = \{1\} \sqcup (0, \delta_0]$, with 1 the physical anchor and $\delta \downarrow 0$ the recovery direction;
3. candidate parameter/operator routes formed from the accepted WDW-5/6 grammar, including formal and failed routes as well as the populated one-rate route;
4. the homogeneous embedding ι_{hom} and a fixed compact tube $K \in U_{\text{hom}}$ containing $(1, 0)$ and the relevant clock segment;
5. phase/amplitude routes formed from the authenticated phase-jet first phase jet, the unit source coefficient, and registered local constructors, including absent, mistyped, formal, failed, and generated routes;
6. the inherited WDW-7 profile clock, physical line, scalar algebra, and complete volume record;
7. the complete continuation coordinate: the WDW-2 proper-reduced regular constraint chart, the WDW-4A spacetime scalar flow, WDW-3 embedding orbit and shifted-tower evaluation map, full characteristic plaque, evaluated clock line, prefix constructor, derived clock-normalizing lapse and canonical characteristic formula, twelve-coordinate RPI role certificate, occupation map, inherited WDW-7 lineage comparator and orbit-identification test, fixed-clock Moore-category type and normalized-seed type; the four baseline/recovery target categories and process functors; the terminal zero exchange monoid, upstream state ledger, four role restrictions, zero history-incidence type, typed target-occupation equalities, separately named closed/enriched lifts, history-indexed exchange cells, strict four-cube type, and downstream target-failure tests;
8. all five residual maps, their fixed seminorms, constants, uniformity range, and first-failure object.

No candidate is required by this definition to succeed. Define $\text{Real}_g^{\text{law}} \subset \text{Cand}_g^{\text{fix}}$ to be the positively populated packages which satisfy every applicable condition with the same fixed coordinates. A new polarization, boundary datum, Hilbert topology, matter factor, positive-frequency selector, nonconstant initial phase curvature, nonconstant initial amplitude, or comparison functional is retained in the ledger as a standing-bearing scope enrichment; it is not silently counted as same-scope and is not declared nonexistent.

Two members are equivalent when they differ only by an orientation-preserving reparameterization of the same vanishing action rate, a phase-frame change $S \mapsto S + \text{constant}$, proper spatial redescription, WDW-3 refoliation skin, or data outside the isolated anchor and asymptotic germ that enter no error, failure, record, continuation, incidence, or comparison coordinate. A different source, clock, classical Cauchy jet, connection, boundary class, phase Hessian seed, amplitude seed, RPI occupation, continuation seed or carrier, target category, process functor, incidence carrier, exchange enrichment, comparison seminorm, error order, or failure locus is standing-bearing.

Proposition 1.6 (Candidate-universe noncircularity). *Membership in $\text{Cand}_g^{\text{fix}}$ does not prescribe the recovered covariant history, the values of the five errors, or the final residue cardinality. It fixes the antecedent source, carrier, permitted constructor grammar, parameter role, first-jet input, domain, quotient, and comparison use before any success is classified.*

Proof. The covariant comparator appears only in the imported handoff. Definition 1.5 does not set the output of the canonical reconstruction map equal to it, does not set any residual to zero, and includes failed and formal routes for first-divergence testing. The phase and amplitude outputs are constructed only later by the independently proved canonical sections of Theorem 1.4. It fixes the types of the continuation coordinates and fixes zero exchange as the declared closed target, but it does not prescribe an RPI occupant, characteristic seed, lineage identity, process arrow, physical target occupation, endpoint equality, or cube verdict. A failed Ward, Hilbert, record, or compatibility test eliminates the candidate rather than changing its ledger. The positive constructions are supplied only in Definition 6.8 and Theorem 6.13. Enriched scopes remain explicit outside the fixed same-scope universe. $\square$

2. Population of the recovery family

2.1 Candidate-parameter exhaustion

The recovery parameter must change the relative weight of quantum differentiation and classical multiplication while leaving the classical target fixed. The accepted corpus already classifies the tempting substitutes.

Candidate	Status	First decisive reason
Renormalization address μ	Eliminated	WDW-6 scale-cylinder transport is the identity at fixed finite normalization; μ is physical skin.
Clock label τ	Eliminated	It indexes relational evolution and does not suppress quantum terms.
Clock momentum ϵ_*	Target change	It changes the occupied Cauchy datum, source history, and energy density.
Classical phase jet	Anchor only	It is a history/Cauchy-jet functional, not the phase of a wavefunctional.
Formal $\hbar \rightarrow 0$	Unpopulated at entry	A symbol without an operator family, states, common domain, and anchor map has no standing.
Born–Oppenheimer mass ratio	Not generated	No gravity–matter factor carrier or invariant polarization is inherited.
Weak-source amplitude	Restricted enrichment	WDW-4 supplies a changing-source family; it is not same-source semiclassical recovery of γ_* .
Response-word order	Error tool only	Truncation controls a source series but does not suppress the quantum principal part.
Single action-rate germ	Survives	It can be applied to the already constructed operator without adding a source, clock, target, or primitive.

Theorem 2.1 (Parameter obstruction and surviving normal form). *Within Definition 1.5, every lawful recovery parameter is equivalent to one positive action-rate coordinate δ with $\delta \downarrow 0$. The scale, clock, history, phase-jet, weak-source, Born–Oppenheimer, and truncation routes do not supply a same-class recovery parameter. A different multi-rate operator deformation is a standing-bearing enriched comparison class.*

Proof. The first four exclusions are the literal role classifications in the WDW-6/7 handoff. A Born–Oppenheimer ratio requires a source-generated factorization and polarization, neither of which is present. Weak-source variation changes C_{src} and the comparator. Word order changes an approximation to one source element while leaving the operator’s quantum/classical balance fixed. For the remaining class, let $s > 0$ multiply the nonmultiplication action scale and vanish in the recovery direction. Any two positive continuous rates with the same orientation are locally related by $s' = f(s)$ after restricting the germ; this is a coordinate change. Scaling distinct inherited primitives at independent rates creates additional subleading data and therefore exits the single-rate class. $\square$

2.2 The constraint-scaled operator family

Let $M_{\mathcal{V}}[N]$ denote multiplication by the complete fixed classical potential coordinate occurring in the established Branch-A Hamiltonian. It is a registered continuous source primitive on $\mathcal{V}_A^{\text{ren}}$. For $0 < \delta \leq \delta_0$ define

$$\hat{\mathcal{H}}_{\delta}[N] := M_{\mathcal{V}}[N] + \delta^2(\hat{\mathcal{H}}^{\text{ren}}[N] - M_{\mathcal{V}}[N]), \quad \varkappa_{\delta} := \delta\hbar. \quad (2)$$

For the linear proper-redescription constraints set

$$\hat{\mathcal{D}}_{\delta}[\xi] := \delta\hat{\mathcal{D}}^{\text{ren}}[\xi]. \quad (3)$$

The physical anchor is not obtained by taking $\delta_0 = 1$ inside the asymptotic proof. It is the separately retained coordinate

$$\hat{\mathcal{H}}_1 = \hat{\mathcal{H}}^{\text{ren}}, \quad \hat{\mathcal{D}}_1 = \hat{\mathcal{D}}^{\text{ren}}, \quad \Psi_1 = F_A^{(0)}.$$

Theorem 2.2 (Common-domain operator population). *Equations (2)–(3) define actual continuous endomorphisms of the accepted graph source $\mathcal{V}_A^{\text{ren}}$ for every $\delta \in \mathbb{P}_8$. All finite registered words have the common source domain $\mathcal{V}_A^{\text{ren}}$ and the common full transpose domain inherited from WDW-6. Every $0 < \delta \leq \delta_0$ is a non-anchor occupant, $\delta = 1$ is the physical anchor, and no regulator or renormalization endpoint is reopened.*

Proof. WDW-6 constructs $\widehat{\mathcal{H}}^{\text{ren}}[N]$, $M_{\mathcal{V}}[N]$, and $\widehat{\mathcal{D}}^{\text{ren}}[\xi]$ as continuous endomorphisms of the completed graph source. A finite linear combination of those actual maps is continuous on the same source. The graph topology contains composition by each primitive; hence every word in the scaled family is continuous on the same completion. The algebraic-transpose extension remains full-domain because it is defined for every continuous source endomorphism. At $\delta = 1$ the formula is the accepted operator. At $\delta \neq 1$ it is a distinct, explicitly enriched comparison operator, not an unregistered interpretation of the physical theory. $\square$

Remark 2.3 (Why the classical principal symbol is fixed). Write the established operator locally as $M_{\mathcal{V}} - \hbar^2 L_2 + \hbar^2 L_{\leq 1}$, with all finite normalization and lower-order terms included in $L_{\leq 1}$ after subtracting the fixed multiplication potential. Acting on Ae^{iS/κ_δ} , the factor $\delta^2 \hbar^2 = \kappa_\delta^2$ cancels the κ_δ^{-2} from two phase derivatives. The leading symbol is therefore the same classical Hamiltonian at every δ ; first-order and zero-order quantum contributions carry at least one positive power of κ_δ .

2.3 The source-generated homogeneous cylinder

Use the two invariant coordinates

$$a(h) = \left(\frac{\text{Vol}_h(\Sigma)}{V_\star} \right)^{1/3}, \quad \varphi(h, \phi) = \text{Vol}_h(\Sigma)^{-1} \int_\Sigma \phi d\mu_h.$$

They are analytic coordinates only. The physical clock remains the function-valued condition $\phi(\cdot) = \tau \mathbf{1}$. Let

$$E_{\text{hom}} f(h, \phi) = f(a(h), \varphi(h, \phi))$$

on a compactly supported cylinder chart and let R_{hom} be restriction along ι_{hom} . Then $R_{\text{hom}} E_{\text{hom}} = \text{id}$. WDW-5 cylinder closure places $E_{\text{hom}} f$ in the smooth trace source; WDW-6 then includes it in $\mathcal{V}_A^{\text{ren}}$.

Let $\mathcal{D}_{\text{hom}} = E_{\text{hom}} C_c^\infty(U_{\text{hom}})$. For a finite lapse-regularity index r and compact $K \Subset U_{\text{hom}}$ define the pulled-back residual seminorm

$$p_{K,r}(F) := \sup_{(a,\varphi) \in K} \sup_{\|N\|_{C^r} \leq 1} |R_{\text{hom}} F[N](a, \varphi)|. \quad (4)$$

This seminorm measures recovery only on the declared homogeneous subtype. It is not a norm on full superspace.

Lemma 2.4 (Compressed Hamiltonian and exact classical symbol). *On $\mathcal{D}_{\text{hom}}$, the pulled-back scaled operator has the form*

$$R_{\text{hom}} \widehat{\mathcal{H}}_\delta E_{\text{hom}} = \mathcal{U}(a, \varphi) - \kappa_\delta^2 \left(G^{AB}(a) \partial_A \partial_B + B^A(a, \varphi) \partial_A + C(a, \varphi) \right), \quad (5)$$

with smooth coefficients on K . Its leading Hamiltonian is

$$\mathcal{H}_{\text{hom}}(a, \varphi, p_a, p_\varphi) = -\frac{\kappa}{12V_\star a} p_a^2 + \frac{1}{2V_\star a^3} p_\varphi^2 + V_\star \left(\frac{3a}{\kappa} + \frac{a^3 m^2 \varphi^2}{2} \right). \quad (6)$$

For

$$p_{a\star} = -\frac{6V_\star c_\star}{\kappa}, \quad p_{\varphi\star} = V_\star \epsilon_\star,$$

one has $\mathcal{H}_{\text{hom}}(1, 0, p_{a\star}, p_{\varphi\star}) = 0$.

Proof. The invariant cylinder has only the volume and homogeneous scalar derivatives. Pulling back the WDW-5/6 second-order contraction therefore gives a two-variable differential operator; the finite normalization, ordering, and counterterm contributions are smooth lower-order coefficients on the compact tube. The classical kinetic coefficients are the ADM reduction of the fixed Einstein–scalar Hamiltonian. For the metric $a^2 h_\star$, curvature is $-6/a^2$ and the compact volume is $V_\star a^3$, which gives (6). Substituting the displayed momenta yields

$$-\frac{3V_\star c_\star^2}{\kappa} + \frac{V_\star \epsilon_\star^2}{2} + \frac{3V_\star}{\kappa} = 0$$

by $c_\star^2 = 1 + \kappa \epsilon_\star^2/6$. $\square$

2.4 Parameter carrier and source attachment

The physical and asymptotic pieces are attached without asserting state convergence. Set

$$\mathbf{a}_8 = (C_{\text{src}}, \mathcal{V}_A^{\text{ren}}, q_\star, \hat{\mathcal{H}}_1 = \hat{\mathcal{H}}^{\text{ren}}, \Psi_1(q_\star) = 1, S(1, 0) = 0, dS(1, 0) = (p_{a\star}, p_{\varphi\star})).$$

Theorem 2.5 (Source-attached comparison-family population). *The operator family of Theorem 2.2, the nonempty homogeneous cylinder of Lemma 2.4, and the attachment tuple $\mathbf{a}_8$ populate the WDW-8 recovery carrier before the covariant comparator is applied. The parameter has a physical role—suppression of every quantum derivative and finite quantum term relative to the fixed classical potential—and is neither gauge, clock, scale skin, source change, nor a blank formal symbol. The five error coordinates are defined on the common seminorm architecture by*

$$(p_{K,r}, d_{\text{Ein}}, p_{I,K}^{\text{cond}}, d_{\text{probe}}, p_{I,K}^{\text{src}}),$$

and are defined before any rate or comparison conclusion is imposed. Hence the recovery family is positively populated.

Proof. There are actual non-anchor operators for all $0 < \delta \leq \delta_0$ and actual cylinder source elements on which they act. The anchor is the exact WDW-6 state/operator coordinate. Source ancestry and classical phase-jet data are fixed by $\mathbf{a}_8$. No occurrence of $H_{4,A}^{\text{cov}}$ enters equations (2), (3), the homogeneous embedding, the seminorm, or the compressed classical Hamiltonian. The remaining seminorms are defined explicitly in the sections which populate their carriers; no rate has yet been assigned. Positive population therefore precedes every recovery comparison. $\square$

3. Hamilton–Jacobi and phase–amplitude recovery

3.1 Noncharacteristic local principal function

Write $q^A = (a, \varphi)$ and let G^{AB} and $\mathcal{U}$ be the coefficients of Lemma 2.4. The Hamilton–Jacobi equation is

$$\mathcal{H}_{\text{hom}}(q, dS) = G^{AB}(q) \partial_A S \partial_B S + \mathcal{U}(q) = 0. \quad (7)$$

On the initial line $\varphi = 0$, apply the source-generated minimal prolongation of Theorem 1.4:

$$S_0(a) = p_{a\star}(a - 1), \quad S_0(1) = 0, \quad S'_0(1) = p_{a\star}. \quad (8)$$

The positive branch of the constraint determines p_φ along that initial line. At $a = 1$ it gives $p_{\varphi\star} = V_\star \epsilon_\star$.

Lemma 3.1 (Noncharacteristic phase population). *After shrinking U_{hom} , equation (7) with initial datum (8) has a unique real smooth solution S on a neighborhood U_S of $(1, 0)$, and*

$$S(1, 0) = 0, \quad dS(1, 0) = (p_{a\star}, p_{\varphi\star}).$$

The positive branch remains separated from $p_\varphi = 0$ on a compact tube $K \Subset U_S$.

Proof. For the Hamiltonian (6),

$$\frac{\partial \mathcal{H}_{\text{hom}}}{\partial p_\varphi} = \frac{p_\varphi}{V_\star a^3},$$

which equals $\epsilon_\star > 0$ at the anchor. Thus the initial hypersurface $\varphi = 0$ is noncharacteristic. The ordinary local method of characteristics for a smooth first-order Hamilton–Jacobi equation gives a unique smooth solution for the prescribed initial function. Continuity and compact restriction preserve a positive lower bound for p_φ . $\square$

Remark 3.2 (First-jet scope). The affine datum is not part of the definition of a successful recovery package. It is the output $P_{\text{ph}}(0, p_{a\star})$ of the unique generated-minimal right inverse constructed before the comparison class is formed. A nonzero second or higher derivative of S_0 is additional standing-bearing Lagrangian data. Such phases may be lawful enriched recovery packages, but they are not silently identified with the present theorem.

3.2 Positive transport amplitude

For the compressed operator (5), define

$$\mathcal{T}_S A := 2G^{AB}(\partial_A A)(\partial_B S) + A(G^{AB}\partial_A \partial_B S + B^A \partial_A S). \quad (9)$$

This includes the accepted first-order ordering coordinate. Apply the second generated-minimal prolongation of Theorem 1.4:

$$A(a, 0) = 1. \quad (10)$$

Lemma 3.3 (Amplitude population). *After a further restriction of K , equation $\mathcal{T}_S A = 0$ with (10) has a unique smooth positive solution. There are constants $0 < c_A < C_A < \infty$ such that*

$$c_A \leq A(q) \leq C_A \quad (q \in K).$$

Proof. The transport vector field is $X_S = 2G^{AB}(\partial_B S)\partial_A$. Its φ component is nonzero on K by Lemma 3.1, so $\varphi = 0$ is also noncharacteristic for the linear transport equation. Integration along its characteristics gives a unique exponential solution. Positive initial data remain positive, and compactness gives the bounds. $\square$

Choose nested compact recovery tubes and a cutoff

$$K \Subset K_+ \Subset U_S, \quad \chi \in C_c^\infty(U_S), \quad \chi \equiv 1 \text{ on a neighborhood of } K_+.$$

For every $0 < \delta \leq \delta_0$ put

$$\psi_\delta^X(q) = \chi(q)A(q) \exp\left(\frac{iS(q)}{\varkappa_\delta}\right), \quad \Psi_\delta = E_{\text{hom}}\psi_\delta^X \in \mathcal{V}_A^{\text{ren}}. \quad (11)$$

The inclusion now follows literally from the declared domain $E_{\text{hom}}C_c^\infty(U_{\text{hom}})$. On K_+ all derivatives of χ vanish, so every local identity on K is identical to the one for the uncut coefficient. We write

$$\psi_\delta := Ae^{iS/\varkappa_\delta} \text{ on } K$$

for that restriction. The cutoff commutator is supported outside K_+ and enters none of the declared recovery seminorms. Thus the family consists of actual smooth cylinder-source elements, while no global constraint solution is claimed for the cutoff extension.

Lemma 3.4 (Cutoff-supported source population). *For every positive recovery rate, Ψ_δ is an actual element of the WDW-5 smooth cylinder source and hence of $\mathcal{V}_A^{\text{ren}}$. The pulled-back action of every registered operator word on K agrees with its action on $Ae^{iS/\varkappa_\delta}$, and the resulting bounds are uniform in δ on the same tube.*

Proof. Compact support and smoothness give the source inclusion. Since χ is identically one on a neighborhood of K_+ , all of its derivatives vanish there. Locality of the finite differential words then removes every cutoff commutator on K . The cutoff is fixed independently of δ , so it does not alter uniformity. $\square$

3.3 Exact substitution identity

Define the quantum remainder

$$\mathcal{Q}A := -A^{-1}(G^{AB}\partial_A \partial_B A + B^A \partial_A A + CA). \quad (12)$$

Theorem 3.5 (Exact WDW phase–amplitude identity). *For every $q \in K$, every $0 < \delta \leq \delta_0$, and every admitted lapse, the actual scaled operator satisfies*

$$A^{-1}e^{-iS/\varkappa_\delta} R_{\text{hom}} \hat{\mathcal{H}}_\delta \Psi_\delta = \mathcal{H}_{\text{hom}}(q, dS) - i\varkappa_\delta A^{-1} \mathcal{T}_S A + \varkappa_\delta^2 \mathcal{Q}A. \quad (13)$$

The first two terms vanish by Lemmas 3.1 and 3.3. No inherited classical phase jet is renamed as S ; S is the solution constructed from the accepted canonical Hamiltonian and its authenticated initial jet.

Proof. Differentiate $Ae^{iS/\varkappa_\delta}$ once and twice in equation (5). The two phase derivatives cancel $\varkappa_\delta^2$ and yield $G^{AB}S_AS_B$. Terms with one phase derivative give exactly $-i\varkappa_\delta\mathcal{T}_SA/A$. Terms with no phase derivative give $\varkappa_\delta^2\mathcal{Q}A$. The multiplication term supplies $\mathcal{U}$. This is an identity of the actual compressed source operator, not a formal truncation. $\square$

The linear constraints vanish on the homogeneous invariant cylinder: $\widehat{\mathcal{D}}_\delta[\xi]\Psi_\delta = 0$. For nonconstant lapse profiles, the leading Hamiltonian density vanishes pointwise on the homogeneous embedding; the remainder estimate is uniform on the declared C^r lapse unit ball.

3.4 Hamilton–Jacobi error and phase attachment

Define

$$E_{\text{HJ}}(\delta) := p_{K,r} \left(A^{-1} e^{-iS/\varkappa_\delta} \widehat{\mathcal{H}}_\delta \Psi_\delta \right). \quad (14)$$

Theorem 3.6 (Uniform Hamilton–Jacobi recovery). *There is a finite constant*

$$C_{\text{HJ}} = \hbar^2 \sup_{q \in K, \|N\|_{C^r} \leq 1} |\mathcal{Q}_N A(q)|$$

such that

$$E_{\text{HJ}}(\delta) \leq C_{\text{HJ}}\delta^2, \quad E_{\text{HJ}}(\delta) \longrightarrow 0.$$

Thus the HJ coordinate is defined, uniformly bounded, and vanishing in the recovery direction; the conclusion follows from the displayed estimate rather than from asymptotic notation alone.

Proof. Equation (13) reduces the residual to $\delta^2\hbar^2\mathcal{Q}A$. The coefficients and A have the finite differentiability required by the seminorm, A is bounded away from zero, the lapse family is bounded in C^r , and K is compact. The displayed supremum is finite and gives the bound. $\square$

Proposition 3.7 (Classical phase-jet attachment). *Under the WDW-1/4 Legendre identification, $dS(1,0)$ is exactly the momentum form of the authenticated WDW-4A normal configuration first jet. Consequently the characteristic of S through the anchor carries the same complete Cauchy datum $z_\star$ and the same source coordinate C_{src} .*

Proof. Lemma 3.1 fixes $p_a = p_{a\star}$ and $p_\varphi = p_{\varphi\star}$. Lemma 2.4 derived those momenta from $K = c_\star h_\star$ and $p_\phi = \epsilon_\star \sqrt{h_\star}$. WDW-1 and WDW-4 prove the regular Legendre map and the exact source lift, so the momentum and velocity versions of the first jet are identical descriptions of the same datum. $\square$

3.5 Maximal recovery domain

Let F_{phase} be the first occurrence of any of:

1. $p_\varphi = 0$ or change of its orientation;
2. failure of the Hamilton–Jacobi characteristic map to remain one-to-one;
3. a caustic or intersection of phase branches;
4. $A = 0$, loss of the positive amplitude bound, or loss of required derivatives;
5. exit from K , the WDW-4A regular chart, or the profile-clock interval;
6. loss of the source, operator, quotient, boundary, or homogeneous-cylinder attachment;
7. divergence of C_{HJ} in the declared topology.

Theorem 3.8 (Local phase–amplitude recovery). *On the maximal component before F_{phase} , the phase–amplitude family is populated, the exact identity (13) holds, and E_{HJ} vanishes uniformly. The theorem selects neither a positive-frequency half-space nor a global WKB branch.*

4. Independent covariant reconstruction and Einstein recovery

4.1 Reconstruction before comparison

The construction order is

$$(S, dS) \mapsto \gamma_S^{\text{can}} \mapsto \text{Rec}_{\text{can} \rightarrow \text{cov}}(\gamma_S^{\text{can}}) \mapsto \text{comparison with } H_{4,A}^{\text{cov}}(\epsilon_\star).$$

The final object is not used in either of the first two arrows.

Let $X_{\mathcal{H}}$ be the Hamilton vector field of (6). The characteristic Lagrangian graph $p = dS$ is invariant under $X_{\mathcal{H}}$ before its first caustic. Let

$$\gamma_S^{\text{hom}}(t) = (a(t), p_a(t); \varphi(t), p_\varphi(t))$$

be the characteristic through $(1, p_{a\star}; 0, p_{\varphi\star})$. Embed it in the full canonical carrier by

$$h(t) = a(t)^2 h_\star, \quad \phi(t) = \varphi(t) \mathbf{1}, \quad (15)$$

and by the Legendre momenta determined by $p_a(t)$ and $p_\varphi(t)$. Choose any positive lapse and compatible shift as presentation data.

Lemma 4.1 (Canonical characteristic history). *The embedded history (15) satisfies the complete Einstein–Klein–Gordon Hamiltonian and momentum constraints and their canonical evolution equations on its maximal local characteristic interval. At the distinguished leaf it equals $z_\star$.*

Proof. Hamilton–Jacobi theory makes every characteristic of a solution to (7) a solution of Hamilton’s equations on the zero-energy surface. Lemma 2.4 is the exact homogeneous reduction of the accepted scalar canonical Hamiltonian, so the embedded variables satisfy the homogeneous full constraints pointwise. Spatial momentum vanishes because the fields are homogeneous and K is proportional to h . Proposition 3.7 gives the initial datum. $\square$

4.2 Local covariant reconstruction

Define

$$\gamma_S^{\text{cov}} := \text{Rec}_{\text{can} \rightarrow \text{cov}}(\gamma_S^{\text{can}})$$

using the WDW-1 source-fibre-preserving prequotient reconstruction map. The Lorentzian metric in one presentation is

$$g_S = -N^2 dt^2 + a(t)^2 h_{\star ij} (dx^i + N^i dt)(dx^j + N^j dt),$$

and the scalar is $\phi_S = \varphi(t)$.

Theorem 4.2 (Independent canonical-to-covariant reconstruction). *The pair (g_S, ϕ_S) is a smooth local Einstein–Klein–Gordon solution with the structured source C_{src} . It is constructed from S , the accepted canonical Hamiltonian, and the WDW-1 reconstruction map before $H_{4,A}^{\text{cov}}(\epsilon_\star)$ is used as a comparator.*

Proof. Lemma 4.1 supplies the constraints and spatial evolution. WDW-1’s ten-component reconstruction theorem and its source-fibre-preserving prequotient equivalence reconstruct all components of the Einstein equation and the Klein–Gordon equation from precisely those canonical data. The WDW-4 pointed lift carries the same source representative, attachment, witness, and Ward coordinate. No comparison with the imported covariant history occurs in these steps. $\square$

4.3 Redescription and refoliation independence

Theorem 4.3 (Slicing independence). *Let two positive lapse–shift presentations cover a common portion of γ_S^{cov} . After WDW-2 proper spatial descent and WDW-3 guarded refoliation descent, their reconstructed covariant objects, source coordinates, relational records, and profile-clock certificate define the same class.*

Proof. WDW-2 proves that proper spatial/internal arrows preserve source and reconstruction ancestry. WDW-3 identifies the regular characteristic directions with the full lapse–shift constraint distribution and proves finite refoliation coherence, source/observable/record preservation, and clock-certificate preservation. The lapse and shift therefore alter only the presentation of the reconstructed history on the guarded overlap. $\square$

The physical clock remains $\phi(\cdot) = \tau \mathbf{1}$. The scalar mean used to define the cylinder is not promoted: on the homogeneous subtype the two numerical values coincide, but only the function-valued level set carries the inherited refoliation transversality theorem.

4.4 Comparison after reconstruction

Choose a compact spacetime slab $\mathcal{K} \subseteq \text{Rec}(K)$ inside the common regular interval. Let d_{Ein} be the quotient-aligned C^{s-2} metric–scalar distance plus the C^{s-3} Einstein–source residual seminorm on $\mathcal{K}$. It is evaluated after minimizing only over the already licensed proper redescription/refoliation arrows.

Theorem 4.4 (Exact Einstein/source comparison). *The independently reconstructed history and the accepted WDW-4A comparator have the same source-attached Cauchy datum. Hence geometric uniqueness gives*

$$[\gamma_S^{\text{cov}}]_{\text{src}} = H_{4,A}^{\text{cov}}(\epsilon_\star)$$

on their common local domain, and

$$E_{\text{Ein}}(\delta) := d_{\text{Ein}}([\gamma_S^{\text{cov}}]_{\text{src}}, H_{4,A}^{\text{cov}}(\epsilon_\star)) = 0 \quad (16)$$

for every $0 < \delta \leq \delta_0$.

Proof. Theorem 4.2 constructs a solution with initial datum $z_\star$ and source C_{src} . The imported WDW-4 history is the geometrically unique maximal globally hyperbolic development of the same datum and source. Cauchy uniqueness gives an initial-surface-fixing isometry, which is a lawful comparison arrow. Equation (16) follows. The comparator has been applied only after the candidate history exists. $\square$

The exact zero is not a claim that the quantum state has no remainder. It says that the leading canonical phase reconstructs the fixed classical history exactly. The quantum amplitude remainder is retained separately in E_{HJ} and E_{back} .

4.5 Connection, transport, and curvature

The WDW-1 target freezes the minimal torsion-free metric branch. Let ∇^S be the derivative reconstructed from g_S .

Theorem 4.5 (Fixed connection and curvature recovery). *On $\mathcal{K}$, ∇^S is the Levi–Civita connection of g_S , belongs to the fixed WDW-1 connection branch, and satisfies*

$$\nabla^S = \nabla^{\gamma_\star}, \quad \text{Riem}(\nabla^S) = \text{Riem}(\nabla^{\gamma_\star})$$

after the comparison isometry. No independent connection, torsion, nonmetricity, tetrad selector, or boundary transport is introduced.

Proof. The reconstructed target is a smooth metric solution on the minimal metric branch. Torsion-free metric compatibility uniquely fixes the Levi–Civita connection. Theorem 4.4 identifies the metrics by an isometry, which transports their Levi–Civita derivatives and curvature tensors exactly. $\square$

4.6 Covariant–canonical recovery

Theorem 4.6 (Exact covariant–canonical recovery). *The canonical-to-covariant constructor exists, is source faithful and slicing independent, and precedes comparison. The Einstein/source and fixed connection/curvature coordinates recover exactly on the declared slab, with $E_{\text{Ein}} = 0$ and with the compact boundaryless class unchanged.*

5. Cauchy-doublet conditional recovery, observables, and records

5.1 The same-scope carrier is not a Born–Oppenheimer factor

Theorem 2.5 establishes before the WKB construction that the fixed handoff contains no gravity–matter tensor factor, invariant polarization, or Born–Oppenheimer mass ratio. The scalar is the inherited profile clock and the only remaining homogeneous configuration coordinate is the gravitational volume coordinate. Consequently a nonzero fixed-background matter Hamiltonian cannot be generated inside the declared scope.

Lemma 5.1 (Conditional-carrier normal form). *On the fixed WDW-8 scope, the only no-spectator conditional carrier available to the recovery germ is the line generated by the complete WDW-7 Cauchy trace of its populated phase–amplitude source element. A point value on one classical characteristic is not an admissible replacement for that trace. Adding a matter tensor factor, polarization, positive-frequency selector, or nonzero fixed-background Hamiltonian is a standing-bearing scope enrichment. Omitting conditional descent violates the same-scope carrier conditions.*

Proof. The WDW-6 physical quotient is positive and rank one. WDW-7 proves that a second-order clock constraint requires the value/normal-response doublet $J_\tau F = (R_\tau F, P_\tau F)$ and eliminates a point-only trace. Theorem 2.1 proves that no factor carrier or polarization is inherited by the comparison family. The fixed-domain alternatives of Theorem 1.3 therefore leave only the generated no-spectator Cauchy-doublet line at the same scope. Each listed additional datum changes the state carrier or operator grammar; absence of the actual J_τ trace violates the required conditional descent. $\square$

5.2 The actual recovered Cauchy doublet

Let $\mathcal{H}_6^{\text{pt}}$ be the accepted WDW-6 physical-radical quotient and let

$$I_6 : \mathcal{H}_6^{\text{pt}} \longrightarrow \mathbb{C}$$

be its canonical isometric coordinate. WDW-7 constructs

$$\text{Cond}_\tau^{\text{C}}[u] = I_6([u])Y_\tau^{(0)}, \quad \mathcal{L}_\tau^{\text{C}} = \mathbb{C}Y_\tau^{(0)}, \quad H_{\text{rel}}^{\text{C}} = 0,$$

where $Y_\tau^{(0)} = J_\tau F_A^{(0)}$ is the quotient-native Cauchy doublet. These are the WDW-7 constructions established in [7, 12].

The scaled constant-normal operator has the exact WDW-7 normal form

$$\hat{\mathcal{C}}_{\delta, N_0} = -\varkappa_\delta^2 (a_\tau \partial_T^2 + B_\tau \partial_T) + \Theta_{\delta, \tau}, \quad \Theta_{\delta, \tau} = M_{\mathcal{V}, \tau} + \delta^2 (\Theta_\tau - M_{\mathcal{V}, \tau}). \quad (17)$$

Thus, before any recovery connection is defined, put

$$\mathbb{A}_{\delta, \tau} = \begin{pmatrix} 0 & I \\ \varkappa_\delta^{-2} a_\tau^{-1} \Theta_{\delta, \tau} & -a_\tau^{-1} B_\tau \end{pmatrix}. \quad (18)$$

The cutoff-supported state Ψ_δ is proper-constraint invariant, so it lies in the domain of the continuous WDW-7 trace. Define its *actual* recovered conditional datum by

$$Y_{\delta, \tau}^{\text{rec}} := J_\tau \Psi_\delta = \begin{pmatrix} R_\tau \Psi_\delta \\ P_\tau \Psi_\delta \end{pmatrix} \in \mathcal{W}_\tau \oplus \mathcal{W}_\tau. \quad (19)$$

At the anchor clock value, the physical evaluation of its first component at $q_\star$ is $A(1, 0) > 0$. Strong graph smoothness therefore gives the nonempty maximal interval

$$I_\delta^{\text{rec}} := \text{Comp}_0\{\tau : Y_{\delta,\tau}^{\text{rec}} \neq 0\}. \quad (20)$$

Set $\mathcal{L}_{\delta,\tau}^{\text{rec}} = \mathbb{C}Y_{\delta,\tau}^{\text{rec}}$. This retains both the value and normal-response components, including the mean-zero germ carried by the parent WDW-7 source.

Define the recovery conditioning map directly on the accepted physical quotient by

$$\text{Cond}_{\delta,\tau}^{\text{rec}}[u] := I_6([u])Y_{\delta,\tau}^{\text{rec}}. \quad (21)$$

Give its range the unique form for which this map is isometric:

$$\langle \text{Cond}_{\delta,\tau}^{\text{rec}} u, \text{Cond}_{\delta,\tau}^{\text{rec}} v \rangle_{\text{rec}} := \langle u, v \rangle_6. \quad (22)$$

Theorem 5.2 (Quotient-native Cauchy-doublet descent). *For every $0 < \delta \leq \delta_0$ and every clock value before the common failure boundary, $\text{Cond}_{\delta,\tau}^{\text{rec}}$ is a unitary isomorphism from $\mathcal{H}_6^{\text{pt}}$ onto $\mathcal{L}_{\delta,\tau}^{\text{rec}}$. It is the unique coefficient-preserving no-spectator map sending the normalized physical class $I_6^{-1}(1)$ to the actual Cauchy datum $J_\tau \Psi_\delta$. Moreover*

$$W_{\delta,\tau} := \text{Cond}_{\delta,\tau}^{\text{rec}}(\text{Cond}_\tau^{\text{C}})^{-1}, \quad W_{\delta,\tau}(\alpha Y_\tau^{(0)}) = \alpha Y_{\delta,\tau}^{\text{rec}}, \quad (23)$$

is a descended unitary intertwiner between the two separately typed lines.

Proof. Surjectivity follows from the definition of the generated range, injectivity from injectivity of I_6 , and isometry from (22). A coefficient-preserving map on a generated one-dimensional fibre is fixed by the image of its normalized generator, proving uniqueness. WDW-7 proves the same statements for $\text{Cond}_\tau^{\text{C}}$; composition gives (23). Equation (19) is formed before this coefficient map and proves that the codomain generator is the complete WKB Cauchy datum itself. Thus the recovery state factors through the physical quotient and the same WDW-7 clock-trace functor. It is neither a point evaluation nor a freely selected abstract line generator. $\square$

5.3 Defect-forced companion transport and conditional equation

Apply R_τ to the actual scaled-constraint residual and define, still before introducing a connection,

$$\mathfrak{f}_{\delta,\tau} := \begin{pmatrix} 0 \\ -\kappa_\delta^{-2} a_\tau^{-1} R_\tau(\widehat{\mathcal{C}}_{\delta,N_0} \Psi_\delta) \end{pmatrix}. \quad (24)$$

The WDW-7 chain rule and the normal form (17) give the exact inhomogeneous companion identity

$$\frac{d}{d\tau} Y_{\delta,\tau}^{\text{rec}} = \mathbb{A}_{\delta,\tau} Y_{\delta,\tau}^{\text{rec}} + \mathfrak{f}_{\delta,\tau}. \quad (25)$$

The term $\mathfrak{f}_{\delta,\tau}$ is fixed by the independently computed order- κ_δ^2 WKB defect. It is not chosen to make a later equation hold. The exact identity of Theorem 3.5, boundedness of a_τ^{-1} on the clock tube, and the fixed cutoff give a finite uniform residual-source seminorm for $\mathfrak{f}_{\delta,\tau}$. No ambient Hilbert norm is inferred. On the generated line define the unique coefficient-linear response map

$$\mathbb{F}_{\delta,\tau}(\alpha Y_{\delta,\tau}^{\text{rec}}) := \alpha \mathfrak{f}_{\delta,\tau}. \quad (26)$$

Now, and only now, define

$$\nabla_{\delta,\tau}^{\text{rec}} \xi := \partial_\tau \xi - \mathbb{A}_{\delta,\tau} \xi - \mathbb{F}_{\delta,\tau} \xi. \quad (27)$$

Although the three terms on the right are evaluated in the ambient Cauchy-germ source, equation (25) proves that their sum lies in $\mathcal{L}_{\delta,\tau}^{\text{rec}}$. For $\xi_{\delta,\tau} = \alpha(\tau) Y_{\delta,\tau}^{\text{rec}}$,

$$\nabla_{\delta,\tau}^{\text{rec}} \xi_{\delta,\tau} = \dot{\alpha}(\tau) Y_{\delta,\tau}^{\text{rec}}. \quad (28)$$

The physical and recovery-line Hamiltonians are separately typed:

$$H_{\text{rel}}^{\text{C}} = 0 \quad \text{on } \mathcal{L}_{\tau}^{\text{C}}, \quad H_{\text{mat},\delta}^{\text{rec}} = 0 \quad \text{on } \mathcal{L}_{\delta,\tau}^{\text{rec}}. \quad (29)$$

The second zero is not stipulated from the first. No gravity–matter split was generated, and after the complete scaled-constraint response has been placed in (27), it is the unique self-adjoint full-domain endomorphism for which coefficient-constant sections satisfy the no-spectator conditional equation.

Theorem 5.3 (Connection and Hamiltonian compatibility). *The connection (27) is Hermitian for (22), is induced from the exact scaled companion identity including its independently formed WKB-defect forcing, and satisfies*

$$\nabla_{\delta,\tau}^{\text{rec}} W_{\delta,\tau} = W_{\delta,\tau} \nabla_{\tau}^{\text{C}}.$$

Consequently $H_{\text{mat},\delta}^{\text{rec}} W_{\delta,\tau} = W_{\delta,\tau} H_{\text{rel}}^{\text{C}}$ exactly. No gravity–matter split, fixed-background operator, or nonzero full-Cauchy Hamiltonian is imported.

Proof. Equation (25) precedes the definition of the connection and fixes both $\mathbb{A}_{\delta,\tau}$ and $\mathbb{F}_{\delta,\tau}$. Equation (28) then says that the recovery connection differentiates only the quotient coefficient. WDW-7 proves the identical coefficient formula for ∇_{τ}^{C} on $\alpha Y_{\tau}^{(0)}$. Equation (23) therefore gives the connection intertwining. Metric compatibility follows by differentiating (22). The Hamiltonian identity follows from the two separately typed zero endomorphisms. The order- $\varkappa_{\delta}^2$ defect remains explicit in $\mathfrak{f}_{\delta,\tau}$ and generates the source-response construction of Section 6. $\square$

For the recovered source-generated Cauchy section $\xi_{\delta,\tau}^{\text{rec}} := \text{Cond}_{\delta,\tau}^{\text{rec}}(I_6^{-1}1) = Y_{\delta,\tau}^{\text{rec}}$ define the actual conditional-equation residual

$$\mathcal{R}_{\text{mat},\delta} := i\varkappa_{\delta} \nabla_{\delta,\tau}^{\text{rec}} \xi_{\delta,\tau}^{\text{rec}} - H_{\text{mat},\delta}^{\text{rec}} \xi_{\delta,\tau}^{\text{rec}}. \quad (30)$$

Thus the conditional matter equation is

$$i\varkappa_{\delta} \nabla_{\delta,\tau}^{\text{rec}} \xi_{\delta,\tau}^{\text{rec}} = H_{\text{mat},\delta}^{\text{rec}} \xi_{\delta,\tau}^{\text{rec}} + \mathcal{R}_{\text{mat},\delta}. \quad (31)$$

For the constructed section both sides vanish for independently established reasons: the exact scaled companion identity, including its displayed forcing, makes the connection term zero, while the fixed-scope Hamiltonian classification gives the zero endomorphism. The forcing is retained visibly in the connection and reappears as the source response below; no two copies of one defect are renamed as opposite matter terms.

Define

$$E_{\text{mat}}(\delta) := \sup_{\tau \in I} \frac{\|\mathcal{R}_{\text{mat},\delta}(\tau)\|_{\text{rec}}}{\varkappa_{\delta} \|\xi_{\delta,\tau}^{\text{rec}}\|_{\text{rec}}}. \quad (32)$$

Theorem 5.4 (Exact profile-clock conditional recovery). *On the declared no-spectator Cauchy-doublet recovery carrier,*

$$E_{\text{mat}}(\delta) = 0 \quad (0 < \delta \leq \delta_0).$$

The result is profile-clock conditioned, physical-quotient descended through the actual $J_{\tau}\Psi_{\delta}$ datum, operator/domain typed, and uniform until the inherited WDW-7 or WKB failure boundary. It does not claim a nonzero matter Hamiltonian on an ungenerated factor carrier.

Proof. Theorem 5.2 supplies physical descent and a nonzero unit generator. Theorem 5.3 and equation (25) give $\mathcal{R}_{\text{mat},\delta} = 0$ identically. The denominator is positive, so the normalized residual is exactly zero on the compact interval. $\square$

5.4 Observable and record comparison

The quantum observable class at the fixed physical scope is $O_{\text{phys}} = \mathbb{C}I$. On every recovery fibre represent it by

$$\rho_{\delta,\tau}(z) = z I_{\mathcal{L}_{\delta,\tau}^{\text{rec}}}.$$

The descended comparison map satisfies

$$W_{\delta,\tau}\rho_\tau^C(z) = \rho_{\delta,\tau}(z)W_{\delta,\tau}.$$

This is exact fixed-algebra intertwining and does not promote a clock-dependent scalar value to one fixed quantum operator.

The complete geometric record is evaluated at the unique profile-clock intersection of the reconstructed history:

$$\mathcal{V}_S^{\text{rec}}(\tau) := \text{Vol}(h_S(\Gamma_S \cap \{\phi = \tau \mathbf{1}\})) = V_\star a_S(\tau)^3. \quad (33)$$

The WDW-7 record is the same construction on $\gamma_\star$.

Theorem 5.5 (Exact observable and complete-record recovery). *On the declared probe class consisting of the represented scalar algebra and the complete volume record,*

$$E_{\text{obs}}(\delta) = 0.$$

Moreover

$$\frac{d}{d\tau}\mathcal{V}_S^{\text{rec}}(0) = \frac{3c_\star}{\epsilon_\star}V_\star > 0.$$

The equality of records descends through proper spatial redescription and WDW-3 refoliation.

Proof. Scalar-algebra intertwining is the displayed descended coefficient identity. Theorem 4.4 identifies the reconstructed and accepted histories after a lawful source-preserving isometry, so their profile-clock intersections and induced metrics agree. Proper spatial transformations preserve total volume, and WDW-3 finite refoliation preserves the complete correlation. At the anchor, $\dot{a}/a = c_\star$ and $\dot{\varphi} = \epsilon_\star$ in unit lapse, hence $da/d\tau = c_\star/\epsilon_\star$. $\square$

Remark 5.6 (Classical record versus quantum observable). The record $\mathcal{V}^{\text{rec}}(\tau)$ is not claimed to be a single operator intertwined between unequal clock values. The fixed quantum algebra is intertwined exactly; the clock-indexed classical complete record is compared as a record. Their types remain distinct.

6. Constraint-tangent backreaction, exact source fibres, and limit scope

6.1 Full-envelope response and attachment-safe source population

The phase–amplitude hierarchy is

$$\begin{aligned} \delta^0 : \mathcal{H}_{\text{hom}}(q, dS) &= 0, \\ \delta^1 : \mathcal{T}_S A &= 0, \\ \delta^2 : r_\delta(q) &:= \mathcal{K}_\delta^2 \mathcal{Q}A(q). \end{aligned}$$

E_{HJ} measures the scalar value of r_δ . Backreaction is the source-extended constrained response generated by that value and its derivatives; it is not a second seminorm on r_δ .

Fix an integer $k \geq 4$, the compact clock interval I , and the slab $\mathcal{K} = I \times \Sigma$. Throughout this section the coordinate on I is the accepted profile clock $t = \tau = \varphi$; N is its positive lapse multiplier. The admitted homogeneous lapse class is

$$\mathcal{N}_{\text{hom}} := \{N \in C^k(I) : 0 < N_- \leq N \leq N_+, \|N\|_{C^k(I)} \leq M_N\}. \quad (34)$$

Every background history γ in the recovery chart has

$$g = -N^2 dt^2 + a^2 h_\star, \quad \phi = \varphi(t), \quad N \in \mathcal{N}_{\text{hom}}.$$

Module A tests the metric witness on the full compact metric-protocol envelope. For that test the accepted scalar-clock representative and its oriented level-set foliation are held fixed; scalar variation is

a different protocol and is not folded into the metric witness. For every nearby metric g' on that fixed foliation set

$$\begin{aligned}\overline{N}_\gamma[g'](t) &:= \frac{1}{\text{Vol}_{g'_t}(\Sigma)} \int_\Sigma N_{g',\gamma}(t, x) d\mu_{g'_t}, \\ a[g'_t] &:= \left(\frac{\text{Vol}_{g'_t}(\Sigma)}{V_\star} \right)^{1/3}.\end{aligned}\tag{35}$$

Here $N_{g',\gamma}$ is the lapse of g' relative to the already fixed γ -clock foliation. Thus (35) is a smooth projection from every compact metric protocol to the declared homogeneous metric coordinates, without varying the clock that defines the slices. Proper redescrptions preserving the oriented clock act naturally on its level sets and leave both averages unchanged. At a homogeneous background a compact spacetime metric variation h has projected components

$$n_h(t) = -\frac{N(t)}{2 \text{Vol}_{g_t}(\Sigma)} \int_\Sigma h(n, n) d\mu_{g_t},\tag{36}$$

$$\alpha_h(t) = \frac{a(t)}{6 \text{Vol}_{g_t}(\Sigma)} \int_\Sigma \text{tr}_{g_t} h^\top d\mu_{g_t},\tag{37}$$

Here n is the future unit normal and $h^\top$ is the spatial restriction. Shift, shear, and mean-zero inhomogeneous variations lie in the recorded kernel of this projection; they are not omitted from the protocol domain.

Define the *metric WDW-defect readout* on that full envelope by

$$\mathfrak{D}_{\delta,\gamma}^g[g'] := \int_I \overline{N}_\gamma[g'](t) r_\delta(a[g'_t], \varphi_\gamma(t)) dt.\tag{38}$$

This readout is formed by applying the registered WDW operator to the registered WKB source element and extracting its exact order- κ_δ^2 coefficient. It is not a scalar action and is never varied in the clock field. Let $\tilde{p}_{\delta,\gamma}$ have as protocols all compactly supported smooth one-parameter metric deformations g_s in this envelope, with the accepted clock representative fixed, and outcome

$$\text{Out}_{\tilde{p}_{\delta,\gamma}}(g_s) := -\left. \frac{d}{ds} \right|_0 \mathfrak{D}_{\delta,\gamma}^g[g_s].\tag{39}$$

The sign is the fixed Module-A metric-witness convention. Denote the complete protocol family by $\mathcal{P}_{\delta,\gamma}$.

Lemma 6.1 (Upstream response-base population). *For every γ in the compact local chart,*

$$\tilde{p}_{\delta,\gamma} = (\mathfrak{D}_{\delta,\gamma}^g, \mathcal{P}_{\delta,\gamma}, \text{Out}_{\tilde{p}_{\delta,\gamma}}, \mathcal{K}, \text{Cont}_{<F_8}, \text{Red}_{\text{pr}})$$

is an admitted response presentation with a complete protocol-response signature. It determines $p_{\delta,\gamma} = [\tilde{p}_{\delta,\gamma}]_{\text{resp}} \in Q_{\text{resp}}$ before metric-source attachment.

Proof. The operator, source element, extraction rule, compact support, and failure status are fixed by the cited dependencies, so the response and its Reference are determinate. Equations (35) and (39) assign an outcome to every admitted compact metric deformation. The projection records both all nonzero lapse/volume responses and its complete shift/shear/mean-zero kernel. Thus the signature is complete relative to the declared envelope rather than selector-generated. Proper redescription preserves total volume, the averaged lapse, the fixed clock representative, the WDW source, and r_δ , hence every outcome. Continuation is the common component before F_8 . Quotienting by equality of these complete signatures gives the stated element of Q_{resp} without using a tensor, action, or source label. $\square$

For a compact metric variation h , define the metric-incidence relation before computing its response image:

$$\text{MetInc}_{p_{\delta,\gamma}}(h, \pi) \iff \begin{cases} \pi = g_s \in \mathcal{P}_{\delta,\gamma}, \\ \dot{g}_0 = h, \\ D\overline{N}_\gamma[h] = n_h, \quad Da[h] = \alpha_h. \end{cases}\tag{40}$$

The complete metric-response relation is

$$\text{MetResp}_{p_{\delta,\gamma}} := \{(h, b) : \exists \pi [\text{MetInc}_{p_{\delta,\gamma}}(h, \pi) \wedge \text{Out}_{p_{\delta,\gamma}}(\pi) = b]\}. \quad (41)$$

At γ , define the continuous functional on the full Module-A metric variation space $\Gamma_c(S^2T^*\mathcal{K})$ by

$$W_{\delta,\gamma}^Q(h) := - \int_I [n_h r_\delta + N(\partial_a r_\delta) \alpha_h] dt. \quad (42)$$

Lemma 6.2 (Module-A attachment before representation). *There is a complete attachment certificate $a_{\delta,\gamma}$ for $p_{\delta,\gamma}$, and*

$$\text{Attach}_g(p_{\delta,\gamma}, a_{\delta,\gamma}, W_{\delta,\gamma}^Q), \quad \text{Graph}(W_{\delta,\gamma}^Q) = \text{MetResp}_{p_{\delta,\gamma}}. \quad (43)$$

Consequently the triple determines an element of the accepted structured Module-A source quotient before an action or stress representative is used.

Proof. Every compact spacetime metric variation has the explicit projections (36)–(37), so the functional is defined on Module A’s complete metric-protocol domain. Variations in the projection kernel have the uniquely forced zero outcome. Provenance is the single readout (38); totality follows from the full protocol family; and the first-variation formula gives representative independence, overlap coherence, additivity, homogeneity, and naturality. Smooth dependence of ADM splitting and volume on the compact C^k envelope, together with the compact C^1 bounds for r_δ , gives LF continuity. Finally, the negative first variation of (38) along every incident metric protocol is exactly (42). Hence the two displayed graphs are equal. Ward closure and action notation are not used in this attachment proof. $\square$

Put $D_t := N^{-1}d/dt$ and $H := D_t a/a$. The lapse and scale coefficients of the already attached metric response give the density and pressure below. Its exchange coefficient is then the unique coefficient which makes the homogeneous Ward identity hold:

$$\rho_\delta^Q[\gamma] := \frac{r_\delta}{V_\star a^3}, \quad p_\delta^Q[\gamma] := -\frac{\partial_a r_\delta}{3V_\star a^2}, \quad J_\delta^Q[\gamma] := \frac{\partial_\varphi r_\delta}{V_\star a^3}. \quad (44)$$

Only after attachment introduce the separate *reduced homogeneous* canonical action

$$\mathcal{J}_\delta^{Q,\text{hom}}[a, \varphi; N] := - \int_I N(t) r_\delta(a(t), \varphi(t)) dt. \quad (45)$$

Its metric first variation agrees with $W_{\delta,\gamma}^Q$ on homogeneous metric protocols, while variation of φ is taken at fixed multiplier N . The full-envelope metric readout (38) is not used as a scalar action. The reduced action therefore gives the canonical forces $-N\partial_a r_\delta$ and $-N\partial_\varphi r_\delta$ without a hidden variation of the clock lapse. Direct differentiation of the three displayed coefficients gives the exact homogeneous balance

$$\mathcal{W}_\delta[\gamma] := D_t \rho_\delta^Q + 3H(\rho_\delta^Q + p_\delta^Q) - J_\delta^Q D_t \varphi = 0. \quad (46)$$

With $\mathcal{J}_{\delta\nu}^Q := -J_\delta^Q \nabla_\nu \phi$, this is $\nabla_\mu T_{Q\nu}^\mu = \mathcal{J}_{\delta\nu}^Q$. The scalar ledger carries $+J_\delta^Q \nabla_\nu \phi$, so the finite composite has zero total internal exchange. All derivatives are the recovered WDW-1 Levi–Civita derivative.

For a represented homogeneous structured source C define the actual source seminorm used below by

$$p_{I,K}^{\text{src}}(C) := \sup_{\substack{h \in \Gamma_c(S^2T^*\mathcal{K}) \\ \|h\|_{C^k} \leq 1}} |W_C(h)| + \|\rho_C\|_{C^{k-1}(I)} + \|p_C\|_{C^{k-1}(I)} + \|J_C\|_{C^{k-1}(I)}. \quad (47)$$

The scalar term is omitted when a component has no scalar-exchange coordinate. This seminorm retains the complete metric witness as well as its homogeneous coordinates; it is not merely a norm of the triplet.

Let $\text{Src}_\varphi^{\text{hom}}(\gamma)$ denote the scalar response, attachment, and tensor coordinates evaluated on the homogeneous history γ . On an undeformed Einstein–Klein–Gordon history it is exactly the WDW-4 map Src_φ . On the response chart it is the direct evaluation of the same independently admitted scalar

action response and attachment, not a fixed old fibre label. In homogeneous coordinates its tensor representative has

$$\rho_\varphi = \frac{1}{2} \left(\frac{p_\varphi^2}{V_\star^2 a^6} + m^2 \varphi^2 \right), \quad p_\varphi^{\text{stress}} = \frac{1}{2} \left(\frac{p_\varphi^2}{V_\star^2 a^6} - m^2 \varphi^2 \right). \quad (48)$$

It is therefore a smooth map on the compact finite-dimensional chart. Its derivative, used below, is the ordinary derivative of the displayed functions; no differentiable structure on WDW-1's full solution quotient is presumed. In particular, for $\zeta = (\zeta_a, \zeta_{p_a}, \zeta_\varphi, \zeta_{p_\varphi})$,

$$\begin{aligned} D\rho_\varphi[\zeta] &= \frac{p_\varphi}{V_\star^2 a^6} \zeta_{p_\varphi} - \frac{3p_\varphi^2}{V_\star^2 a^7} \zeta_a + m^2 \varphi \zeta_\varphi, \\ Dp_\varphi^{\text{stress}}[\zeta] &= \frac{p_\varphi}{V_\star^2 a^6} \zeta_{p_\varphi} - \frac{3p_\varphi^2}{V_\star^2 a^7} \zeta_a - m^2 \varphi \zeta_\varphi. \end{aligned} \quad (49)$$

The derivative of its complete metric witness is obtained by differentiating the same imported scalar-action first variation; its coefficients are the displayed smooth functions. Thus the entire structured-source derivative, not only its density and pressure coordinates, is constructed on this chart.

For a homogeneous solution history γ of the deformed model, set

$$C_{\text{tot}}^\epsilon[\gamma] := \text{Src}_\varphi^{\text{hom}}(\gamma) \boxplus \epsilon C_\delta^Q[\gamma], \quad C_\delta^Q[\gamma] := [p_{\delta,\gamma}, a_{\delta,\gamma}, W_{\delta,\gamma}^Q]_{\text{src}}. \quad (50)$$

Here $\boxplus$ is the accepted finite Module-A composite. At the declared homogeneous scope the reduced Einstein–Klein–Gordon action is augmented by $\epsilon \mathcal{S}_\delta^{Q,\text{hom}}$. Its reduced phase carrier and symplectic form are unchanged, and its constraint is $\Phi_{\text{hom}} + \epsilon r_\delta$. No full field-theoretic WDW-1 strong lift is asserted for the deformed family.

Lemma 6.3 (History-indexed source and reduced-action population). *For every sufficiently small positive δ , every $|\epsilon| \leq 1$, and every homogeneous solution history γ in the compact chart, $C_\delta^Q[\gamma]$ is an actual smooth Module-A structured source and $C_{\text{tot}}^\epsilon[\gamma]$ is an attachment-certified finite composite. The reduced action (45) has Hamiltonian contribution r_δ , zero momentum-density contribution, and exactly the scale and scalar forces displayed above. Moreover*

$$p_{I,K}^{\text{src}}(C_\delta^Q[\gamma]) \leq C_{\text{src}} \delta^2$$

uniformly on the chart.

Proof. Lemmas 6.1 and 6.2 establish source admission and graph equality before representation. Direct variation of the separate reduced action at fixed N gives its Hamiltonian projection, zero homogeneous momentum projection, and the two stated forces. Equation (46) supplies the component exchange, while the reduced scalar Euler–Lagrange identity supplies the opposite term; hence the homogeneous composite is on the closed Ward locus. Since A is smooth on a neighbourhood of K , compact coefficient bounds give $\|r_\delta\|_{C^{k+1}(K)} \leq C_{r,k} \delta^2$. Every term in (42), (44), and the derivatives entering (47) therefore has the stated order. $\square$

6.2 A homogeneous right inverse and exact history-dependent source family

Let Φ^ϕ denote the WDW-4 coupled scalar constraint map on the accepted regular chart, and let $\mathfrak{e}_{\text{hom}} : Z_{\text{hom}} \hookrightarrow Z_{\text{can}}$ be the exact homogeneous canonical embedding. The full-space WDW-4 right inverse is not presumed to preserve this subcarrier. On $Z_{\text{hom}} = T^*U_S$ instead put

$$\Phi_{\text{hom}} = \mathcal{H}_{\text{hom}}, \quad v_\varphi(z) = \partial_{p_\varphi} \mathcal{H}_{\text{hom}}(z) = \frac{p_\varphi}{V_\star a^3} > 0$$

and define

$$Q_z^{\text{hom}}(f) := \frac{f}{v_\varphi(z)} \partial_{p_\varphi} \in T_z Z_{\text{hom}}. \quad (51)$$

Then

$$D\Phi_{\text{hom},z}Q_z^{\text{hom}}(f) = f, \quad D\Phi_{\text{hom},z}^\phi D\mathfrak{e}_{\text{hom},z}Q_z^{\text{hom}}(f) = \iota_{\mathcal{H}}f. \quad (52)$$

The second identity is the exact homogeneous reduction; its momentum component is zero. WDW-4 remains load-bearing for the surrounding regular constraint and source-family chart, not for an invariant-subspace property of its pseudodifferential inverse [4].

Deform the constraint by

$$\Phi_{\delta,\text{hom}}^\epsilon := \Phi_{\text{hom}} + \epsilon r_\delta$$

and use $N(t)\Phi_{\delta,\text{hom}}^\epsilon$ as its coordinate-time generator. Thus lapse is a presentation coordinate, not part of the constraint surface. Set $\widehat{r}_\delta = r_\delta(1,0)$. After shrinking δ_0 ,

$$p_{\varphi,\delta}^\epsilon := \sqrt{p_{\varphi\star}^2 - 2V_\star\widehat{r}_\delta} \quad (53)$$

is real for $|\epsilon| \leq 1$. Keeping $(a, \varphi, p_a) = (1, 0, p_{a\star})$ and replacing $p_{\varphi\star}$ by this root gives $z_\delta^\epsilon(0) \in Z_{\text{hom}}$ with

$$\Phi_{\text{hom}}(z_\delta^\epsilon(0)) + \epsilon r_\delta(1,0) = 0, \quad z_\delta^0(0) = z_\star. \quad (54)$$

Its derivative is

$$\zeta_{\delta,N}(0) := \partial_\epsilon z_\delta^\epsilon(0)|_0 = -Q_{z_\star}^{\text{hom}}(r_\delta(z_\star)) \in T_{z_\star}Z_{\text{hom}}, \quad (55)$$

so $D\Phi_{\text{hom},z_\star}\zeta_{\delta,N}(0) + r_\delta(z_\star) = 0$.

Let $z_{\delta,N}^\epsilon(\tau)$ be the profile-clock normalization of the Hamiltonian evolution generated by $N(t)(\Phi_{\text{hom}} + \epsilon r_\delta)$. Its added canonical columns are exactly

$$\dot{p}_a|_Q = -\epsilon N \partial_a r_\delta, \quad \dot{p}_\varphi|_Q = -\epsilon N \partial_\varphi r_\delta, \quad \mathcal{H}|_Q = \epsilon r_\delta, \quad \mathcal{H}_i|_Q = 0. \quad (56)$$

These are the ADM constraint and evolution projections of the already attached source (44); hence the canonical system and the source system coincide in every ADM column, not merely in the initial constraint and Ward identity. Since $\{\Phi_{\delta,\text{hom}}^\epsilon, N\Phi_{\delta,\text{hom}}^\epsilon\} = 0$, the exact constraint is preserved. The vector field is tangent to Z_{hom} , so the whole family remains in that carrier.

The reduced equations themselves provide the covariant/source check; no new WDW-1 strong lift is inserted. Define

$$\rho_{\text{tot}} := \rho_\varphi + \epsilon \rho_\delta^Q, \quad p_{\text{tot}} := p_\varphi^{\text{stress}} + \epsilon p_\delta^Q. \quad (57)$$

Using $p_a = -6V_\star a^2 H/\kappa$ and $p_\varphi = V_\star a^3 D_t \varphi$, the constraint and the four Hamilton equations are algebraically equivalent to

$$H^2 - \frac{1}{a^2} = \frac{\kappa}{3} \rho_{\text{tot}}, \quad (58)$$

$$D_t H + \frac{1}{a^2} = -\frac{\kappa}{2} (\rho_{\text{tot}} + p_{\text{tot}}), \quad (59)$$

$$D_t^2 \varphi + 3H D_t \varphi + m^2 \varphi = -\epsilon J_\delta^Q. \quad (60)$$

The momentum equation is identically zero on the homogeneous cylinder. These are respectively the normal constraint, spatial Einstein evolution, and sourced Klein–Gordon equation for $g = -N^2 dt^2 + a^2 h_\star$ and the exact source C_{tot}^ϵ . Multiplying (60) by $D_t \varphi$ gives

$$D_t \rho_\varphi + 3H(\rho_\varphi + p_\varphi^{\text{stress}}) = -\epsilon J_\delta^Q D_t \varphi, \quad (61)$$

which cancels ϵ times (46). Thus the complete homogeneous Einstein and scalar equations, not selected initial columns, are verified directly.

For each exact history $\gamma_{\delta,N}^\epsilon := (z_{\delta,N}^\epsilon, N)$, assign the exact history-dependent source

$$C_{\delta,N}^\epsilon := \text{Src}_\varphi^{\text{hom}}(\gamma_{\delta,N}^\epsilon) \boxplus \epsilon C_\delta^Q[\gamma_{\delta,N}^\epsilon]. \quad (62)$$

At $\epsilon = 0$ this is the inherited source C_{src} . For $\epsilon \neq 0$ it is not the old fixed source label: the scalar source is reevaluated on the changed momentum and history. Its tangent is

$$\dot{C}_{\delta,N}^{\text{tot}} := \partial_\epsilon C_{\delta,N}^\epsilon \Big|_0 = D \text{Src}_\varphi^{\text{hom}}(\gamma_\star)[\zeta_{\delta,N}] + C_\delta^Q[\gamma_\star]. \quad (63)$$

This is the complete source tangent; the first term may not be dropped.

Write $v_\varphi = X_{\mathcal{H}}\varphi$ and $V_S = v_\varphi^{-1}X_{\mathcal{H}}$. With Poisson tensor J on Z_{hom} set

$$X_{r_\delta} := J dr_\delta. \quad (64)$$

Because r_δ has no momentum dependence, $X_{r_\delta}\varphi = 0$. Division by Nv_φ cancels the lapse exactly and gives

$$B_{\delta,N}^{\text{clk}} := \frac{1}{v_\varphi} \left(X_{r_\delta} - \frac{X_{r_\delta}\varphi}{v_\varphi} X_{\mathcal{H}} \right), \quad B_{\delta,N}^{\text{clk}}\varphi = 0. \quad (65)$$

The subscript retains the coordinate-time presentation; the displayed force is independent of it. Differentiating the exact family at $\epsilon = 0$ gives

$$\frac{d\zeta_{\delta,N}}{d\tau} = DV_S(z_S(\tau))\zeta_{\delta,N} + B_{\delta,N}^{\text{clk}}(z_S(\tau)), \quad (66)$$

with initial condition (55). Differentiation of the preserved constraint gives on every clock slice

$$\mathcal{C}_{\delta,N}^{(1)}(\tau) := D\Phi_{\text{hom},z_S(\tau)}\zeta_{\delta,N}(\tau) + r_\delta(z_S(\tau)) = 0. \quad (67)$$

6.3 Explicit local reconstruction and the backreaction topology

No derivative is taken on WDW-1's full solution quotient. Define instead the local homogeneous history chart

$$\mathcal{X}_{I,K}^k := \{(z, N) : z \in C^k(I, K), N \in \mathcal{N}_{\text{hom}}, \varphi = \tau\} \quad (68)$$

and the explicit reconstruction

$$\mathfrak{R}_{\text{hom}}(z, N, C) := (-N^2 dt^2 + a^2 h_\star, \varphi, C). \quad (69)$$

The scalar structured-source coordinate in this chart is the smooth explicit map (48); the quantum coordinate is the first-variation map (42). Therefore $\mathfrak{R}_{\text{hom}}$ and the complete source assignment (62) are C^1 between the stated finite C^k charts. Define the covariant response directly by

$$\begin{aligned} \eta_{\delta,N} &:= \partial_\epsilon \Big|_0 \mathfrak{R}_{\text{hom}}(z_{\delta,N}^\epsilon, N, C_{\delta,N}^\epsilon) \\ &= (2a \zeta_a h_\star, \zeta_\varphi, \dot{C}_{\delta,N}^{\text{tot}}). \end{aligned} \quad (70)$$

Its norm is

$$\|\eta_{\delta,N}\|_{\text{rec}} := \|2a\zeta_a h_\star\|_{C^{k-1}(\mathcal{X})} + \|\zeta_\varphi\|_{C^{k-1}(I)} + p_{I,K}^{\text{src}}(\dot{C}_{\delta,N}^{\text{tot}}). \quad (71)$$

On the compact chart there is a constant $C_{\mathfrak{R}}$ such that

$$\|\eta_{\delta,N}\|_{\text{rec}} \leq C_{\mathfrak{R}}(\|\zeta_{\delta,N}\|_{C^{k-1}(I)} + p_{I,K}^{\text{src}}(C_\delta^Q[\gamma_\star])). \quad (72)$$

This follows by differentiating the displayed algebraic reconstruction and source formulas. At $\epsilon = 0$, WDW-1's accepted pointed scalar lift and Theorem W1.5 identify this chart with the inherited source-attached covariant history. For $\epsilon \neq 0$, equations (58)–(61) prove the declared homogeneous covariant statement directly. No full deformed WDW-1 source-family lift, full Dirac ledger, or global solution-manifold structure is claimed.

Theorem 6.4 (Direct homogeneous backreaction population). *For every sufficiently small $\delta > 0$ and every $N \in \mathcal{N}_{\text{hom}}$, equations (53)–(62) populate an exact homogeneous source-extended constrained family. Its source fibre is the history-dependent $C_{\delta,N}^\epsilon$, its canonical and source ADM projections agree in every homogeneous constraint, metric-evolution, and scalar-evolution column. Its explicit FLRW reconstruction satisfies (58)–(61) exactly. Its unique linear response is (66), its complete source tangent is (63), and its explicitly reconstructed covariant response is (70).*

Proof. Equations (53)–(54) prove exact initial constraint occupation in Z_{hom} . Equations (51)–(52) prove both inversion and carrier preservation. Equation (56) identifies the complete reduced Hamilton equations with the homogeneous ADM projections of the attached source. Substitution of the two Legendre identities gives (58)–(60), and (46) plus (61) gives exact total conservation. These displayed equations establish the complete homogeneous covariant/source match directly. Smooth dependence of finite-dimensional ODE solutions yields (66); differentiating the exact constraint and exact source map gives (67) and (63). Finally (69) differentiates directly to (70). No fixed old scalar fibre, full-space tangent inverse, deformed WDW-1 strong lift, or derivative of the WDW-1 quotient is used. $\square$

Let

$$\mathcal{N}_{\delta,N} := z_{\delta,N}^1 - z_S - \zeta_{\delta,N}. \quad (73)$$

Smooth parameter dependence and $\|r_\delta\|_{C^{k+1}(K)} = O(\delta^2)$ give

$$\|\mathcal{N}_{\delta,N}\|_{C^0(I)} \leq C_{\text{nl}}\delta^4. \quad (74)$$

Define the component and total exchange residuals by

$$\Xi_{\delta,N}^Q := \nabla_\mu T_{Q\nu}^\mu + J_\delta^Q \nabla_\nu \phi, \quad (75)$$

$$\Xi_{\delta,N}^{\text{tot}} := \nabla_\mu (T_{\varphi\nu}^\mu + \epsilon T_{Q\nu}^\mu). \quad (76)$$

The first is zero by (46); the scalar equation supplies the opposite current, so the second is zero on every exact history. In E_{back} , $\Xi_{\delta,N}^{\text{tot}}$ denotes the $\epsilon = 1$ member; the identity is uniform for $|\epsilon| \leq 1$.

Define the complete backreaction error by

$$\begin{aligned} E_{\text{back}}(\delta) := & \sup_{N \in \mathcal{N}_{\text{hom}}} (\|\zeta_{\delta,N}\|_{C^{k-1}(I)} + \|\eta_{\delta,N}\|_{\text{rec}} + p_{I,K}^{\text{src}}(\dot{C}_{\delta,N}^{\text{tot}}) + p_{I,K}^{\text{src}}(C_\delta^Q[\gamma_\star]) \\ & + \|\mathcal{N}_{\delta,N}\|_{C^0(I)} + \|\mathcal{C}_{\delta,N}^{(1)}\|_{C^{k-2}(I)} \\ & + \|\mathcal{W}_\delta\|_{C^{k-2}(I)} + \|\Xi_{\delta,N}^Q\|_{C^{k-2}(\mathcal{K})} \\ & + \|\Xi_{\delta,N}^{\text{tot}}\|_{C^{k-2}(\mathcal{K})}). \end{aligned} \quad (77)$$

Theorem 6.5 (Backreaction and source-exchange estimate). *There is a finite constant C_{back} , independent of δ and the admitted lapse presentation, such that*

$$E_{\text{back}}(\delta) \leq C_{\text{back}}\delta^2 \longrightarrow 0.$$

The linearized constraint, component Ward, and total exchange residuals vanish exactly; the canonical, explicit covariant, and complete source responses are $O(\delta^2)$, and the nonlinear remainder is $O(\delta^4)$.

Proof. The explicit inverse and initial datum give $\|\zeta(0)\| \leq C_Q\delta^2$. Equations (64) and (65) give a uniform $O(\delta^2)$ forcing on the compact tube. Variation of constants and Gronwall’s inequality give $\|\zeta\|_{C^{k-1}} \leq C_I\delta^2$ uniformly over $\mathcal{N}_{\text{hom}}$; lapse independence is the exact cancellation in (65). The explicit derivatives of (48) and the estimate in Lemma 6.3 show from (63) that $p_{I,K}^{\text{src}}(\dot{C}^{\text{tot}}) = O(\delta^2)$. Equation (72) gives the same order for η without differentiating the WDW-1 reconstruction equivalence. Equation (74) is higher order, and (67), (46), (75), and (76) make the remaining terms zero. $\square$

Remark 6.6 (Separation from the HJ and conditional coordinates). A locally constant r_δ contributes to E_{HJ} but has no local Hamiltonian force; it still produces the homogeneous initial constraint correction in (55). In the conditional analysis the same defect appears as the independently computed forcing in the Cauchy companion connection. Thus E_{HJ} measures the scalar defect, E_{mat} tests the source-corrected no-spectator conditional equation, and E_{back} measures the complete history-dependent source response, explicit covariant reconstruction, Ward balance, and nonlinear remainder.

6.4 Same-source lineage and the Module-C specialization

Theorem 6.7 (Same-source Ward/exchange recovery). *The undeformed characteristic, reconstructed Einstein equation, conditional record, and response family meet at the one inherited source $C_{\delta,N}^0 = C_{\text{src}}$. For $\epsilon \neq 0$ the exact source is the history-dependent class (62), not the old fibre label. Its tangent is the sum in (63). The additional component is generated by the WDW-defect response, admitted by graph attachment, and represented only afterward by the Module-A composite. All gravitational and scalar exchange terms cancel using the recovered Levi–Civita connection.*

Proof. At $\epsilon = 0$, the history is the accepted WDW-4 history and the explicit scalar map agrees with Src_φ , giving C_{src} . For nonzero ϵ , equations (48) and (62) re-evaluate the scalar tensor on the changed p_φ and complete history. Differentiation therefore gives both terms in (63). Lemmas 6.1–6.3 construct and admit the quantum response from the same WDW defect. Equations (46) and (76) prove internal and total exchange closure. $\square$

Module C is parametric over a populated regular local fixed-clock packet that discharges its upstream certificate. Its established vacuum/full-Hilbert example is not that certificate and is not imported here. The WDW-8 packet is instead formed from two finite predecessor constructions whose analytic roles must not be conflated: WDW-4A’s ordinary spacetime scalar flow and WDW-3’s shifted-tower evaluation equivalence from Cauchy embeddings to the canonical characteristic plaque. The derivative-losing ADM constraint field is never integrated as an ordinary vector field on a fixed Sobolev phase space. Its clock-normalized canonical formula is recovered only after the evaluated embedding path has already been constructed.

Let $\pi_{\text{pr}} : \mathcal{C}_{\text{reg}} \rightarrow \mathcal{Q}_{\text{reg}}^{\text{pr}}$ be the accepted WDW-2 proper-spatial quotient of the regular coupled constraint surface. Within the fixed compact hyperbolic source branch, let $\mathcal{S}_{8,\phi}^{\text{reg}} \subset \mathcal{Q}_{\text{reg}}^{\text{pr}}$ be the guarded profile-clock submanifold represented by

$$D_i\phi = 0, \quad \phi = \tau \mathbf{1}, \quad p_\phi/\sqrt{h} > c_0 > 0, \quad \tau \in I_8 \subseteq I_7, \quad (78)$$

with the fixed source, attachment, branch, WDW-6 state, and WDW-7 prefix coordinates retained and with F_7 not reached. The constant value in (78) defines a smooth clock readout $\tau : \mathcal{S}_{8,\phi}^{\text{reg}} \rightarrow I_8$. WDW-4A characteristic transversality and its function-valued implicit-function theorem make this a split regular local submanifold of the accepted WDW-2 chart, not a scalar-mean slice.

Definition 6.8 (Evaluated-plaque upstream packet and RPI occupation). Let $(M_{\gamma_*}, g_{\gamma_*}, \phi_{\gamma_*})$ and $i_* : \Sigma \hookrightarrow M_{\gamma_*}$ be the WDW-4A history and anchored Cauchy embedding. On its certified clock neighborhood put

$$V_\phi := \frac{\nabla\phi_{\gamma_*}}{g_{\gamma_*}^{-1}(d\phi_{\gamma_*}, d\phi_{\gamma_*})}, \quad V_\phi(\phi_{\gamma_*}) = 1, \quad F_t^\phi := \text{Fl}_t^{V_\phi}. \quad (79)$$

This is the established smooth spacetime flow from WDW-4A, restricted to one common short interval before its first clock failure. WDW-3 supplies a shrunk smooth-tame embedding orbit $\mathcal{O}_8^{\text{emb}}$ about i_* and its evaluation equivalence

$$\text{Eval}_{\gamma_*} : \mathcal{O}_8^{\text{emb}} \xrightarrow[\text{Nash} - \text{Moser}]{\cong} \mathcal{P}_8 \subset \mathcal{C}_{\text{reg}}, \quad (80)$$

where $\mathcal{P}_8$ is the contractible characteristic plaque. Define the actual clock-forward embedding path and its evaluated proper-quotient line by

$$i_u := F_{u-u_*}^\phi \circ i_*, \quad (81)$$

$$\Phi_8^{\text{pl}} : I_8^\circ \longrightarrow \mathcal{P}_8^{\phi, \text{pr}} \subset \mathcal{S}_{8,\phi}^{\text{reg}}, \quad \Phi_8^{\text{pl}}(u) := \pi_{\text{pr}}(\text{Eval}_{\gamma_*}(i_u)). \quad (82)$$

After shrinking once, every i_u lies in $\mathcal{O}_8^{\text{emb}}$ and $\tau \circ \Phi_8^{\text{pl}} = \text{id}_{I_8^\circ}$; hence Φ_8^{pl} is a smooth chart of its full occupied continuation image. The upstream seed and prefix are constructed by

$$\gamma_8^{\text{pl}}(\Phi_8^{\text{pl}}(u)) := d\Phi_{8,u}^{\text{pl}}(\partial_u), \quad (83)$$

$$r_8^{\text{pl}}(\Phi_8^{\text{pl}}(u)) := [v \longmapsto \Phi_8^{\text{pl}}(v)]_{v \in [u_-, u], \text{pref}}. \quad (84)$$

Only now differentiate the already formed embedding path. Since its leaves are scalar level sets,

$$\dot{i}_u = V_\phi \circ i_u = N_\phi n_u, \quad N_\phi = \frac{1}{n_u(\phi)} = \frac{\sqrt{h}}{p_\phi}. \quad (85)$$

WDW-3 differentiated evaluation gives the derived proper-quotient identity

$$\mathcal{V}_8^{\text{pl}} = d\pi_{\text{pr}}(X_{(N_\phi, 0)}) \quad (86)$$

modulo its specified proper-spatial shift convention. Thus the canonical characteristic formula records the tangent of a path already constructed by spacetime flow and evaluation; it is not used to generate a fixed-stage ODE. Independently of every downstream target, declare the terminal closed-exchange monoids and their unique restriction maps

$$\mathcal{J}_{D,8}^{\text{cl}} := \mathbf{0}_D = (\{0_D\}, \oplus, 0_D), \quad \mathcal{J}_{X,8}^{\text{cl}} := \mathbf{0}_X, \quad \rho_X : \mathbf{0}_D \longrightarrow \mathbf{0}_X, \quad \rho_X(0_D) = 0_X, \quad (87)$$

for $X \in \{g, H, \text{rec}, \text{comp}\}$. The upstream state ledger on the evaluated packet is the constant map

$$\ell_{D,8}^{\text{pl}} : \mathcal{P}_8^{\phi, \text{pr}} \longrightarrow \mathbf{0}_D, \quad \ell_{D,8}^{\text{pl}}(\Phi_8^{\text{pl}}(u)) := 0_D, \quad \ell_{X,8}^{\text{pl}} := \rho_X \circ \ell_{D,8}^{\text{pl}}. \quad (88)$$

This fixes the same-scope terminal exchange coordinate before occupation, projection, or target testing. It does not assert that a proposed physical process is closed: failure of any later Ward, Hilbert, record, or compatibility test means that process does not inhabit this declared packet; the ledger remains fixed throughout the test. Thus the WDW-3 full plaque is present as the regular neighborhood, while the physical continuation is the WDW-4A-generated clock line within it. Neither object is obtained by integrating the canonical ADM field, and no target projection has yet been defined. The upstream Module-C role signature is the following literal assignment, not a list of desired downstream outputs.

Role coordinate	Upstream WDW-4/6/7 realization
Σ_{loc}	The compact carrier Σ , WDW-3 embedding orbit $\mathcal{O}_8^{\text{emb}}$, full evaluated plaque $\mathcal{P}_8$, and occupied line $\mathcal{P}_8^{\phi, \text{pr}}$.
Σ_{trans}	The WDW-4A spacetime seed V_ϕ , its embedding path i_u , and the evaluated seed $\mathcal{V}_8^{\text{pl}}$.
Σ_{field}	The complete canonical field packet $z = (h, \pi; \phi, p_\phi)$ with attached C_{src} .
Σ_{part}	The non-null scalar occupant $p_\phi/\sqrt{h} > c_0$ and $\rho_\phi > 0$.
Σ_{gauge}	The WDW-2 proper quotient, WDW-3 guarded characteristic groupoid, and Q_ϕ .
Σ_{resp}	WDW-3 differentiated evaluation of $\dot{i}_u$ and its derived canonical response $\delta_{(N_\phi, 0)}$, with the accepted scalar source attachment.
Σ_{vac}	The authenticated nonzero WDW-6 physical line, state $F_A^{(0)}$, and coordinate I_6 ; no vacuum metric is assumed.
Σ_{clock}	The complete WDW-4A/7 package $D_{\text{clk}}^{(7)}$, its flow K_ϕ , orientation, and readout τ .
Σ_{rec}	The constructed prefix $r_8^{\text{pl}}(\Phi_8^{\text{pl}}(u))$, its complete metric-volume readout, and order $\preceq_7$; these are identified with ρ_u and $\mathcal{V}^{\text{rec}}(u)$ only afterward.
Σ_{comp}	The generated WDW-7 compatibility quotient and its no-transport verdict.
Σ_{fail}	The inherited clock boundary and first-failure object F_7 .
Σ_{exch}	The independently declared closed packet $(\mathbf{0}_D, \ell_{D,8}^{\text{pl}}, \{\rho_X : \mathbf{0}_D \rightarrow \mathbf{0}_X\}_{X \in \{g, H, \text{rec}, \text{comp}\}})$ of (87)–(88).

Write this assignment as

$$\Sigma_I^8(\Phi_8^{\text{pl}}(u)) = (\Sigma_{\text{loc}}, \Sigma_{\text{trans}}, \Sigma_{\text{field}}, \Sigma_{\text{part}}, \Sigma_{\text{gauge}}, \Sigma_{\text{resp}}, \Sigma_{\text{vac}}, \Sigma_{\text{clock}}, \Sigma_{\text{rec}}, \Sigma_{\text{comp}}, \Sigma_{\text{fail}}, \Sigma_{\text{exch}}). \quad (89)$$

Let D_8^{pl} be the realized-persistence regime carried by these packets and let $I_{D_8^{\text{pl}}}$ be the lawful RPI interior supplied after the antecedents below are verified. The occupation map is

$$\mathcal{J}_8 : \mathcal{P}_8^{\phi, \text{pr}} \longrightarrow I_{D_8^{\text{pl}}}, \quad \mathcal{J}_8(\Phi_8^{\text{pl}}(u)) := [\Sigma_I^8(\Phi_8^{\text{pl}}(u))]_{\text{RPI}}. \quad (90)$$

The unique RPI state ledger on the full lawful interior is

$$\ell_{D,8}^{\text{cl}} : I_{D_8^{\text{pl}}} \longrightarrow \mathbf{0}_D, \quad \ell_{D,8}^{\text{cl}}(x) := 0_D, \quad \ell_{D,8}^{\text{cl}} \circ j_8 = \ell_{D,8}^{\text{pl}}. \quad (91)$$

On the occupied image define the RPI-level seed

$$\mathcal{U}_8^{\text{up}}(j_8(\Phi_8^{\text{pl}}(u))) := d_{j_8, \Phi_8^{\text{pl}}(u)}(\gamma_8^{\text{pl}}(\Phi_8^{\text{pl}}(u))). \quad (92)$$

No WDW-8 recovered development, downstream Hilbert flow, Hamiltonian, F_8 , nonzero incidence, or recovery parameter occurs in (79)–(92).

Lemma 6.9 (Module-C antecedents, actual lineage, and seed exhaustion). *The packet D_8^{pl} satisfies the complete Module-C upstream certificate: local Lipschitz regularity, clock normalization, standing tangency, nonzero seed, fail-closed prefixes, formation before projection, and exact inherited record and compatibility preservation. It also satisfies*

$$K_{D_8^{\text{pl}}} \wedge \text{FixedAdmRealPers}_{D_8^{\text{pl}}} \wedge \text{RoleSeq}_{D_8^{\text{pl}}}.$$

The already constructed evaluated-plaque translation

$$\alpha_t^{\text{pl}}(\Phi_8^{\text{pl}}(u)) := \Phi_8^{\text{pl}}(u + t) \quad (93)$$

therefore descends through the occupation by

$$\alpha_t^8 \circ j_8 = j_8 \circ \alpha_t^{\text{pl}}, \quad (94)$$

and Module C supplies its nonempty fixed-clock Moore category $\text{Cont}_{D,8}^{\tau}$. For the authenticated anchor $x_{\star} = j_8(\Phi_8^{\text{pl}}(u_{\star}))$, its occupied objects are

$$x_8(u) := \alpha_{u-u_{\star}}^8(x_{\star}) = j_8(s_u), \quad s_u = [\Gamma_{\star} \cap \{\phi(\cdot) = u\mathbf{1}\}]_{\text{pr}}, \quad (95)$$

and its retained prefix is $r_8^{\text{up}}(u) := [x_8|_{[u-,u]}]_{\text{pref}} = \rho_u$. Positive-duration arrows f_{u_0,u_1} are nonidentity and satisfy

$$f_{u_1,u_2} \circ f_{u_0,u_1} = f_{u_0,u_2}, \quad f_{u,u} = 1_{x_8(u)}. \quad (96)$$

The zero state ledger descends to the occupied RPI objects, and the history incidence formed immediately after the Moore category is

$$J_{D,8}^{\text{cl}} : \text{Mor}(\text{Cont}_{D,8}^{\tau}) \longrightarrow \mathbf{0}_D, \quad J_{D,8}^{\text{cl}}(f) := 0_D. \quad (97)$$

It satisfies, before any target is evaluated,

$$\ell_{D,8}^{\text{cl}}(x_1) = J_{D,8}^{\text{cl}}(f) \oplus \ell_{D,8}^{\text{cl}}(x_0), \quad f : x_0 \rightarrow x_1, \quad (98)$$

$$J_{D,8}^{\text{cl}}(1_x) = 0_D, \quad J_{D,8}^{\text{cl}}(f_2 \circ f_1) = J_{D,8}^{\text{cl}}(f_2) \oplus J_{D,8}^{\text{cl}}(f_1). \quad (99)$$

Finally, $\mathcal{U}_8^{\text{up}}$ is the unique same-scope characteristic seed on the declared quotient that is clock normalized and preserves the fixed source, branch, clock-use, record, and failure coordinates.

Proof. The WDW-4A timelike-gradient theorem constructs the ordinary smooth spacetime flow F_t^{ϕ} on one common compact clock neighborhood. Composition with the fixed embedding gives the smooth path $u \mapsto i_u$ before canonical evaluation. WDW-3 Theorem W3.5 maps every such guarded embedding path to a characteristic path, and its W3.6 Nash–Moser theorem makes (80) a smooth-tame local equivalence with the declared fixed derivative shift. No same-index phase-space inverse-function or ODE theorem is used.

In the explicit u -chart on $\mathcal{P}_8^{\phi,\text{pr}}$, γ_8^{pl} is the constant vector ∂_u and hence is locally Lipschitz with Lipschitz constant zero. Its translation (93) is already defined and satisfies the local flow laws directly. The accepted spacetime scalar equation gives

$$d\tau(\mathcal{U}_8^{\text{up}})\mathbf{1} = V_{\phi}(\phi)\mathbf{1} = \mathbf{1}. \quad (100)$$

Thus the seed is nonzero and (84) assigns an actual coherent prefix to every point of the occupied line. The retained canonical datum, clock readout, and prefix separate the local packet, so j_8 is a regular embedding onto its image and (92) is well typed.

Equations (85)–(86) prove tangency to the coupled constraint surface and, because $V_\phi(\phi) = 1$ spatially, to the profile condition $D_i\phi = 0$. The fixed source, attachment, branch, proper quotient, WDW-6 state, prefix order, and F_7 guards are the accepted WDW-4/6/7 persistence data. Restriction before their first failure makes every prefix fail closed, and the accepted WDW-7 record and compatibility quotients advance along that same guarded characteristic segment. This proves the seven Module-C clauses directly.

Kernel necessity applies to the displayed nondegenerate determinate packet, giving $K_{D_8^{\text{pl}}}$. Fixed source and branch, non-null scalar response, regular quotient rank, bounded local carrier, oriented clock, retained prefix, and explicit first failure give $\text{FixedAdmRealPers}_{D_8^{\text{pl}}}$. Every coordinate of $\text{RoleSeq}_{D_8^{\text{pl}}}$ has the independently formed realization displayed in the table preceding (89); Σ_{exch} carries the actual zero state ledger and its four unique role restrictions. The RPI theorem therefore supplies the coherent closed interior and makes (90) lawful. The accepted parametric Module-C theorem and its upstream results 33.1–33.5 [14] now apply to the independently formed seed and translation. Equation (94) defines their lawful RPI descent; Module C gives nonemptiness, standing preservation, prefix closure, and (96) before any target functor is evaluated. Because the source and target ledgers of every history take the unique value 0_D , equations (97)–(99) follow from the zero-monoid law. They are the independently registered Module-C Definition 22.2 certificate; no Ward, Hilbert, record, or compatibility output has been used to form them.

The path $\Phi_8^{\text{pl}}(u)$ lies on the WDW-3 characteristic plaque by Theorem W3.5 and has scalar value $u\mathbf{1}$ by (100). WDW-4A gives a unique guarded intersection of the occupied orbit with $\phi(\cdot) = u\mathbf{1}$; WDW-7 denotes exactly that proper-quotient point by s_u . Hence the first equality in (95) is an orbit-identification theorem, not the definition of the seed. WDW-7's complete-record descent then gives $r_8^{\text{up}}(u) = \rho_u$ and $\mathcal{V}^{\text{rec}}(u) = \text{Vol}(h(s_u))$.

For uniqueness, any same-scope RPI-level characteristic seed has a representative $X_{(N,X)}$. On $D_i\phi = 0$, clock normalization requires

$$1 = \mathcal{L}_X\phi + N \frac{p\phi}{\sqrt{h}} = N \frac{p\phi}{\sqrt{h}},$$

so $N = N_\phi$. The remaining X is a proper-spatial direction and $d\pi_{\text{pr}}(X_{(0,X)}) = 0$. The descended seed is therefore exactly $\mathcal{U}_8^{\text{up}}$. A noncharacteristic transverse component changes the continuation/response carrier; a component changing source, branch, record, or failure data changes an explicitly retained coordinate. Either is a standing-bearing enrichment, not a second same-scope seed. This proves complete characteristic-seed exhaustion without assuming a constant residual flow-box coordinate. $\square$

For $u \in I_8$ let $z_u := z(x_8(u))$ be the complete canonical datum carried by the upstream object, let $C_{\text{src},u} := C_{\text{src}}(x_8(u))$, and let $I_{6,u} := I_6(x_8(u))$. These are evaluations of retained upstream coordinates, not objects reconstructed from the clock label. For $\bullet \in \{0, \delta\}$ write

$$G_u^0 := (g_u^0, K_u^{\text{geom},0}, \varphi_u^0, p_{\varphi,u}^0; C_{\text{src},u}^0), \quad (101)$$

$$G_u^\delta := (g_u^\delta, K_u^{\text{geom},\delta}, \varphi_u^\delta, p_{\varphi,u}^\delta; C_{\text{src},u}^\delta), \quad (102)$$

$$\mathcal{L}_u^0 := \mathcal{L}_u^C, \quad \mathcal{L}_u^\delta := \mathcal{L}_{\delta,u}^{\text{rec}}, \quad (103)$$

$$\mathcal{V}_u^0 := \mathcal{V}^{\text{rec}}(u), \quad \mathcal{V}_u^\delta := \mathcal{V}_\delta^{\text{rec}}(u), \quad (104)$$

$$K_u^\bullet := [x_8(u), I_{6,u}, C_{\text{src},u}, \phi(z_u) = u\mathbf{1}, \mathcal{C}_u^\bullet = 0, \mathcal{W}_u^\bullet = 0, \quad (105)$$

$$\mathcal{T}_u^\bullet = 0, F_8^0 > u, F_8^\delta > u]_{\text{comp}}. \quad (106)$$

Here 0 denotes the accepted WDW-4/7 baseline and δ the independently reconstructed WDW-8 recovery object. The symbols $\mathcal{C}^\bullet$, $\mathcal{W}^\bullet$, and $\mathcal{T}^\bullet$ are, respectively, the exact constraint, total Ward, and hidden-source-transfer residuals. Thus K_u^δ is an explicitly defined object, not an undeclared notation. The target failure objects $F_8^\bullet$ are formed from the fixed downstream tests of Section 9; they are not coordinates of

the upstream packet. After both target families have been evaluated, define

$$I_8^{\text{elig}} := \{u \in I_8 : F_8^0 > u \text{ and } F_8^\delta > u\}, \quad \text{Cont}_{D,8}^{\tau,\text{elig}} := \text{Cont}_{D,8}^\tau|_{\{x_8(u):u \in I_8^{\text{elig}}\}}. \quad (107)$$

This is a downstream eligibility restriction of an already formed upstream category, not a source of its standing. The exact baseline and recovery theorems make I_8^{elig} an open nonempty anchor neighborhood; shrink only the reported target slab inside it. These profiles are not those of Module C’s previously exhibited vacuum/full-Hilbert inhabitant.

Role	Constructed baseline/recovery objects	Completeness data
Metric/source	G_u^0, G_u^δ	Nonvacuum density $\epsilon_*^2/2 > 0$; WDW-4 source attachment and WDW-8 exact reconstruction, with gauge, regularity, compact boundary class and Cauchy data.
Quantum line	$\mathbb{P}(\mathcal{L}_u^0), \mathbb{P}(\mathcal{L}_u^\delta)$	Full projective-ray sets of the actual one-dimensional carriers, with transition probability, Hermitian transport, generator graph/domain and source ledger specified below.
Record	$\mathcal{V}_u^0, \mathcal{V}_u^\delta$	Separate baseline/recovery quotient objects with fixation, lineage, proper/refoliation descent, reuse and failure status.
Compatibility	K_u^0, K_u^δ	Separate final quotients carrying source, clock, boundary, exact constraint, Ward and no-hidden-transfer coordinates.

Definition 6.10 (WDW-8 role target categories). For $u \in I_8^{\text{elig}}$, let $\text{Met}_{D,8}$ have the objects G_u^0, G_u^δ and the lawful source-attached history restrictions as arrows, modulo the inherited WDW-2/3 presentation equivalence. Let $\text{Hilb}_{D,8}$ have the full ray sets of $\mathcal{L}_u^0$ and $\mathcal{L}_u^\delta$ as objects and the projectivizations of their unitary parallel transports as arrows. Let $\text{Rec}_{D,8}$ have the separate baseline/recovery record classes as objects and oriented retained-lineage segments as arrows. Let $\text{Comp}_{D,8}$ have the separately defined classes K_u^0, K_u^δ as objects and their quotient-preserving continuations as arrows. In every case the zero-length segment is the identity and endpoint-compatible concatenation is composition.

The quantum row is defined by explicit carrier data, not by dimension alone. Put $Y_u^0 := Y_u^{(0)}$ and $Y_u^\delta := Y_{\delta,u}^{\text{rec}}$, the already constructed nonzero source-generated sections. For each of the two line bundles define

$$\begin{aligned} \mathbf{h}_u^\bullet &:= (\mathbb{P}(\mathcal{L}_u^\bullet), [Y_u^\bullet], p([v], [w]) = |\langle v, w \rangle|^2 / (\|v\|^2 \|w\|^2), \\ &\quad P_{u_1, u_0}^\bullet, \text{Graph}(0) = \mathcal{L}_u^\bullet \times \{0\}, \\ &\quad \text{Dom}(0) = \mathcal{L}_u^\bullet, C_{\text{src}}, \sigma_{\text{line}}). \end{aligned} \quad (108)$$

where $\bullet \in \{0, \delta\}$ and $P^\bullet$ is parallel transport for ∇^C or $\nabla_\delta^{\text{rec}}$. Each projective set contains exactly one ray because the underlying nonzero carrier is one-dimensional; it is nevertheless the entire ray set of that carrier. The transports are unitary and strongly continuous, their parallel-frame Stone generator is the full-domain zero operator, and their graph, transported nonzero section, and source ancestry are displayed in (108). In particular,

$$Y_{u_1}^\bullet = P_{u_1, u_0}^\bullet Y_{u_0}^\bullet, \quad \nabla_{\partial_u}^{H, \bullet} Y_u^\bullet = 0. \quad (109)$$

The coordinate σ_{line} is the explicit spatial/source ledger

$$\sigma_{\text{line}} := (\mathcal{W}_u \oplus \mathcal{W}_u, C_{\text{src}}, J_u, \text{TRACE-GENERATED-LINE, NO-FULL-CAUCHY-PROMOTION}). \quad (110)$$

It records the actual WDW-7 Cauchy-germ source, ancestry and trace map and states that this is the source-generated clock-line target, not Module C’s unrelated spatial full-Cauchy Schrödinger target.

Lemma 6.11 (Target-specific fibre completeness). *The four categories of Definition 6.10 are independently complete on the displayed WDW-8 targets. In $\text{Met}_{D,8}$, lawful equivalence is fixed by the complete source system, attachment, gauge, regularity, boundary and Cauchy profile. In $\text{Hilb}_{D,8}$ it is fixed by (108)–(110). In $\text{Rec}_{D,8}$ it is fixed by the complete WDW-7 quotient, fixation, lineage, reuse and failure profile. In $\text{Comp}_{D,8}$ it is fixed by the complete joint-standing, source, clock, constraint, Ward, boundary and no-transport profile. Equality of one selected readout is never substituted for equality of these profiles.*

Proof. WDW-4 and Theorems 4.2–4.4 supply the exact nonvacuum source-attached Cauchy profile and WDW-2/3 supply its lawful orbit equivalence. Section 5 constructs both nonzero line carriers, their forms, full ray sets, connections, unitary transports, zero generator graphs and domains, trace/source ledgers and failure boundary. WDW-7 Theorems 18.1 and 30.1 prove finality, descent and lineage for $\mathcal{V}^{\text{rec}}$. The compatibility object is the final quotient by the complete displayed coordinates, so its arrows preserve every one of them. These four conclusions prove the target-specific form of Module-C Theorem 34.3 directly; they do not appeal to the profile of Module C’s earlier vacuum/full-Hilbert inhabitant. $\square$

Definition 6.12 (Baseline and recovery process functors). For $f = f_{u_0, u_1}$ in $\text{Cont}_{D,8}^{\tau, \text{elig}}$ define, after $\text{Cont}_{D,8}^{\tau}$ has been formed,

$$\beta_8^{g,0}(f) := \text{Dev}^0(f), \quad \beta_8^{g,\delta}(f) := \text{Dev}^\delta(f), \quad (111)$$

$$\beta_8^{H,0}(f) := \mathbb{P}(P_{u_1, u_0}^C), \quad \beta_8^{H,\delta}(f) := \mathbb{P}(P_{\delta; u_1, u_0}^{\text{rec}}), \quad (112)$$

$$\beta_8^{\text{rec},0}(f) := [\mathcal{V}^{\text{rec}}|_{[u_0, u_1]}]_{\text{lin}}, \quad \beta_8^{\text{rec},\delta}(f) := [\mathcal{V}_\delta^{\text{rec}}|_{[u_0, u_1]}]_{\text{lin}}, \quad (113)$$

$$\beta_8^{\text{comp},0}(f) := [K_{u_0}^0 \rightsquigarrow K_{u_1}^0]_{\text{comp}}, \quad \beta_8^{\text{comp},\delta}(f) := [K_{u_0}^\delta \rightsquigarrow K_{u_1}^\delta]_{\text{comp}}. \quad (114)$$

Their object maps are the corresponding endpoint objects in the table. Denote the resulting baseline and recovery functors by $\Pi_8^{X,0}$ and $\Pi_8^{X,\delta}$. For a general upstream object x write $z(x)$ for its retained canonical datum, $C_{\text{src}}(x)$ for its attached source, $\tau(x)$ for its physical profile-clock value, $I_6(x)$ for its WDW-6 coordinate, $F_A^{(0)}(x)$ for its authenticated state coordinate, and $r(x)$ for its inherited prefix. The object maps factor through that complete state by the explicit constructors

$$\Pi_8^{g,\bullet}(x) := \text{Dev}^\bullet(z(x), C_{\text{src}}(x)) = G_x^\bullet, \quad (115)$$

$$\Pi_8^{H,\bullet}(x) := \mathbb{P}\left(\text{Cond}_{\tau(x)}^\bullet(I_6(F_A^{(0)}(x)))\right) = \mathbb{P}(\mathcal{L}_x^\bullet), \quad (116)$$

$$\Pi_8^{\text{rec},\bullet}(x) := \text{Rec}(\text{Dev}^\bullet(z(x), C_{\text{src}}(x)), \tau(x), r(x)) = \mathcal{V}_x^\bullet, \quad (117)$$

$$\Pi_8^{\text{comp},\bullet}(x) := \text{JointStand}(x, I_6(x), C_{\text{src}}(x), \tau(x), r(x), \mathcal{C}^\bullet, \mathcal{W}^\bullet, \mathcal{J}^\bullet, F_8^0, F_8^\delta) = K_x^\bullet. \quad (118)$$

At $x = x_8(u)$ these objects are precisely $G_u^\bullet, \mathbb{P}(\mathcal{L}_u^\bullet), \mathcal{V}_u^\bullet, K_u^\bullet$. The arrow $\text{Dev}^\bullet(f)$ is defined explicitly as

$$\text{Dev}^\bullet(f_{u_0, u_1}) := [u \mapsto \text{Dev}^\bullet(z(x_8(u)), C_{\text{src}}(x_8(u)))]_{u \in [u_0, u_1]}. \quad (119)$$

Cauchy uniqueness makes the endpoint restrictions agree. Thus the clock label does not synchronize an independently solved history to an arbitrary curve: every target is a function of the complete upstream state, and the metric arrow is the induced process of that state segment. Their seed-compatibility equations are

$$d\Pi_8^{g,\bullet}(\mathcal{U}_8^{\text{up}}) = \partial_u \gamma_u^{g,\bullet}, \quad \nabla_{\partial_u}^{H,\bullet} Y_u^\bullet = 0, \quad (120)$$

$$d\Pi_8^{\text{rec},\bullet}(\mathcal{U}_8^{\text{up}}) = \dot{\mathcal{V}}^\bullet(u), \quad d\Pi_8^{\text{comp},\bullet}(\mathcal{U}_8^{\text{up}}) = \partial_u K_u^\bullet, \quad (121)$$

where $\bullet \in \{0, \delta\}$. The Hilbert equation acts on the transported nonzero section, and its projective class gives the derivative of the ray object. Prefix retention is carried by the constructed class $r_8^{\text{up}}(u)$ in (95); the four downstream values are generated by the displayed object maps and are not pulled back to define the RPI seed.

Here $\gamma_\delta^{\text{rec}}$ is the independently reconstructed history of Theorem 4.2, not the backreaction family used to define E_{back} . Theorem 4.4 gives $\gamma_\delta^{\text{rec}} = \gamma_\star^{\text{cov}}$ lawfully on the recovery domain, and Theorem 5.5 gives $\mathcal{V}_\delta^{\text{rec}} = \mathcal{V}^{\text{rec}}$. The compatibility object uses those same exact source, clock, constraint, Ward and failure coordinates.

For composable subsegments the processes satisfy the literal equations

$$\beta_8^{g,\bullet}(f_{12}) \circ \beta_8^{g,\bullet}(f_{01}) = \beta_8^{g,\bullet}(f_{02}), \quad \mathbb{P}(P_{u_2,u_1}^\bullet) \mathbb{P}(P_{u_1,u_0}^\bullet) = \mathbb{P}(P_{u_2,u_0}^\bullet), \quad (122)$$

$$\beta_8^{\text{rec},\bullet}(f_{12}) \circ \beta_8^{\text{rec},\bullet}(f_{01}) = \beta_8^{\text{rec},\bullet}(f_{02}), \quad \beta_8^{\text{comp},\bullet}(f_{12}) \circ \beta_8^{\text{comp},\bullet}(f_{01}) = \beta_8^{\text{comp},\bullet}(f_{02}), \quad (123)$$

with identities at zero duration. Thus the symbols β_8^X are actual target-category arrows with proved identity and composition laws.

The exact reconstruction, record, and compatibility theorems construct lawful equivalences

$$\text{Eq}_{\delta,u}^g : G_u^0 \longrightarrow G_u^\delta, \quad \text{Eq}_{\delta,u}^{\text{rec}} : \mathcal{V}_u^0 \longrightarrow \mathcal{V}_u^\delta, \quad \text{Eq}_{\delta,u}^{\text{comp}} : K_u^0 \longrightarrow K_u^\delta. \quad (124)$$

These are arrows between separately constructed complete objects; their existence is not part of either target definition. Define endpoint comparison maps

$$\Theta_{\delta,u}^g := \text{Eq}_{\delta,u}^g, \quad \Theta_{\delta,u}^H := \mathbb{P}(W_{\delta,u}), \quad \Theta_{\delta,u}^{\text{rec}} := \text{Eq}_{\delta,u}^{\text{rec}}, \quad \Theta_{\delta,u}^{\text{comp}} := \text{Eq}_{\delta,u}^{\text{comp}}. \quad (125)$$

The metric arrow is licensed by exact reconstruction, the quantum arrow is the projectivization of the constructed unitary (23), and the other two use only the separately proved complete-profile equivalences (124).

The terminal exchange coordinate has already been registered upstream by (87)–(98). The following equations test whether the independently constructed targets inhabit that closed coordinate; they do not define or modify it. Set $\nabla^{H,0} := \nabla^C$, $\nabla^{H,\delta} := \nabla_\delta^{\text{rec}}$, $H_{\text{rel}}^0 := H_{\text{rel}}^C$, and $H_{\text{rel}}^\delta := H_{\text{mat},\delta}^{\text{rec}}$, all already constructed in Section 5. For each complete target let $\pi_{\text{src}}^{X,\bullet}$ denote its specified source-ledger projection and $\beta_{8,\text{src}}^{X,\bullet}(f)$ the corresponding component of the actual process arrow. Define the typed Hilbert occupation readout only on the transported nonzero section,

$$\mathcal{H}_u^\bullet := \left(\nabla_{\partial_u}^{H,\bullet} Y_u^\bullet, H_{\text{rel}}^\bullet Y_u^\bullet \right) \in \mathcal{L}_u^\bullet \oplus \mathcal{L}_u^\bullet. \quad (126)$$

For every $f = f_{u_0,u_1}$ the four closed-occupation conditions are

$$\begin{aligned} \mathcal{W}_u^\bullet &= 0, & \mathcal{H}_u^\bullet &= (0,0), \\ \pi_{\text{src}}^{\text{rec},\bullet} \Pi_8^{\text{rec},\bullet}(x_8(u_1)) &= \beta_{8,\text{src}}^{\text{rec},\bullet}(f) (\pi_{\text{src}}^{\text{rec},\bullet} \Pi_8^{\text{rec},\bullet}(x_8(u_0))), \\ \pi_{\text{src}}^{\text{comp},\bullet} \Pi_8^{\text{comp},\bullet}(x_8(u_1)) &= \beta_{8,\text{src}}^{\text{comp},\bullet}(f) (\pi_{\text{src}}^{\text{comp},\bullet} \Pi_8^{\text{comp},\bullet}(x_8(u_0))). \end{aligned} \quad (127)$$

The last two lines are equalities in their own final quotient fibres; no subtraction, additive completion, common anchor fibre, or representative choice is introduced.

Now form Module C's split carrier on the already declared zero role monoid,

$$\tilde{Y}_{8,\text{cl}}^{X,\bullet} := Y_8^{X,\bullet} \times \mathbf{0}_X. \quad (128)$$

The closed projection, its lifted predictor, and the actual enriched process are separately named but coincide in this closed specialization:

$$\tilde{\Pi}_{8,\text{cl}}^{X,\bullet}(x) := (\Pi_8^{X,\bullet}(x), 0_X), \quad (129)$$

$$\hat{\beta}_{8,0}^{X,\bullet}(f)(y, 0_X) := (\beta_8^{X,\bullet}(f)y, 0_X), \quad \tilde{\beta}_{8,\text{cl}}^{X,\bullet}(f) := \hat{\beta}_{8,0}^{X,\bullet}(f). \quad (130)$$

Identity and composition follow from the already proved base-process laws and the unique zero-monoid action:

$$\tilde{\beta}_{8,\text{cl}}^{X,\bullet}(f_2) \circ \tilde{\beta}_{8,\text{cl}}^{X,\bullet}(f_1) = \tilde{\beta}_{8,\text{cl}}^{X,\bullet}(f_2 \circ f_1), \quad \tilde{\beta}_{8,\text{cl}}^{X,\bullet}(1_x) = 1. \quad (131)$$

For each history $f : x_0 \rightarrow x_1$, rather than for each object, define the Module-C predicted and actual endpoints

$$z_8^{0,X,\bullet}(f) := \widehat{\beta}_{8,0}^{X,\bullet}(f) \widetilde{\Pi}_{8,\text{cl}}^{X,\bullet}(x_0), \quad (132)$$

$$z_8^{1,X,\bullet}(f) := \widetilde{\Pi}_{8,\text{cl}}^{X,\bullet}(x_1) = \widetilde{\beta}_{8,\text{cl}}^{X,\bullet}(f) \widetilde{\Pi}_{8,\text{cl}}^{X,\bullet}(x_0). \quad (133)$$

The base-process endpoint equation and $\rho_X J_{D,8}^{\text{cl}}(f) = 0_X$ give

$$z_8^{0,X,\bullet}(f) = z_8^{1,X,\bullet}(f). \quad (134)$$

Hence the directed exchange cell required by Module C Definition 22.5 is the history-indexed identity

$$E_8^{X,\bullet}(f) : z_8^{0,X,\bullet}(f) \xrightarrow{0_X} z_8^{1,X,\bullet}(f), \quad E_8^{X,\bullet}(f) = 1_{z_8^{0,X,\bullet}(f)}. \quad (135)$$

It satisfies the source-target-compatible composition law

$$E_8^{X,\bullet}(f_2 \circ f_1) = E_8^{X,\bullet}(f_2) \circ \widehat{\beta}_{8,0}^{X,\bullet}(f_2)(E_8^{X,\bullet}(f_1)). \quad (136)$$

For every history f and $\varepsilon = (\varepsilon_g, \varepsilon_H, \varepsilon_{\text{rec}}, \varepsilon_{\text{comp}}) \in \{0, 1\}^4$, set

$$\mathcal{K}_{8,f}^{\bullet}(\varepsilon) := \prod_{X \in \{g, H, \text{rec}, \text{comp}\}} z_8^{\varepsilon_X, X, \bullet}(f). \quad (137)$$

An edge in direction X is (135) whiskered by identities in the other three factors. All sixteen vertices are therefore constructed with the correct history source, every face and monotone path commutes strictly, and (136) makes the cube assignment functorial under history concatenation.

Theorem 6.13 (Module-C specialization and recovery naturality). *The RPI occupation (90) lawfully inhabits the accepted Module-C upstream type. The category $\mathbf{Cont}_{D,8}^T$ is nonempty and contains positive-duration nonidentity histories before target evaluation; its anchor orbit is exactly the WDW-7 occupied lineage (95). Its later eligible full subcategory (107) is also nonempty and contains positive-duration histories. The eight processes in (111) define baseline and recovery functors on that same already formed eligible subcategory, and each object and arrow factors through the complete upstream state as in (115). For every f_{u_0, u_1} and every $X \in \{g, H, \text{rec}, \text{comp}\}$,*

$$\Theta_{\delta, u_1}^X \circ \beta_8^{X,0}(f) = \beta_8^{X,\delta}(f) \circ \Theta_{\delta, u_0}^X. \quad (138)$$

The equations paste under composition. The upstream RPI state ledger and history incidence (91) and (97) satisfy Module-C Definition 22.2 before target evaluation. Both baseline and recovery targets satisfy the four independent closed-occupation tests (127). Consequently the separately typed data (128)–(137) define, for every history, baseline and recovery sixteen-vertex strict closed cubes with identity edges and functorial history composition. The backreaction amplitude ϵ is not used as an upstream clock or incidence coordinate.

Proof. The antecedent certificate, RPI occupation, nonemptiness, actual-lineage identification, nonidentity, and category laws are proved by Lemma 6.9 before any WDW-8 target is evaluated. Exact baseline occupation and the local recovery theorems make both target-failure predicates false at the anchor and on a common open neighborhood, proving the eligible subcategory nonempty without backflow into upstream standing. A one-sided shrinking inside that neighborhood is unnecessary: the closed history certificate is defined on the whole eligible subcategory. Segment restriction and concatenation give the metric equations in (122); Hermitian connection transport gives the quantum equation; concatenation of one fixed record lineage gives the record equation; and final-quotient preservation gives the compatibility equation. Equations (115) and (120) prove full-state factorization and smooth projectability from the independently formed seed. These prove functoriality rather than synchronizing solved histories by their common clock label.

For $X = g$, Theorems 4.2 and 4.4 construct the two complete nonvacuum source-attached developments, so both sides of (138) are the same restricted history. For $X = H$, Theorem 5.3 states $\nabla_\delta^{\text{rec}} W_\delta = W_\delta \nabla^C$; uniqueness of parallel transport gives

$$\mathbb{P}(W_{\delta,u_1})\mathbb{P}(P_{u_1,u_0}^C) = \mathbb{P}(P_{\delta;u_1,u_0}^{\text{rec}})\mathbb{P}(W_{\delta,u_0}).$$

For $X = \text{rec}$, Theorem 5.5 constructs equality of the complete clock-indexed record paths, including descent and lineage. For $X = \text{comp}$, the exact source, clock, constraint, Ward and target-failure coordinates construct $\text{Eq}_{\delta,u}^{\text{comp}}$, so the two final-quotient arrows intertwine. Applying these equations twice and using (122) proves pasting.

Finally, baseline WDW-4 source conservation and the direct homogeneous Ward theorem give $\mathcal{W}^0 = \mathcal{W}^\delta = 0$. Hermitian parallel transport gives $\nabla_{\partial_u}^{H,\bullet} Y_u^\bullet = 0$, while the separately typed full-domain zero generators give $H_{\text{rel}}^\bullet Y_u^\bullet = 0$. Thus the well-typed section readout (126) gives $\mathcal{H}^0 = \mathcal{H}^\delta = (0,0)$. WDW-7 record descent says exactly that the endpoint source coordinate is the image of the incoming coordinate under the retained-lineage arrow, proving the record equality in (127). The separately constructed compatibility no-transport theorem proves the compatibility equality there. These four typed results verify that both completed physical target packages inhabit the independently declared zero exchange coordinate; none is used to manufacture that coordinate.

The base-process endpoint law together with the already registered identity $\ell_{D,8}^{\text{cl}}(x_1) = J_{D,8}^{\text{cl}}(f) \oplus \ell_{D,8}^{\text{cl}}(x_0)$ proves (134). Thus every directed role cell is the required history-indexed identity, not an objectwise anchor cell. Whiskered coordinatewise pasting gives the strict cube $\mathcal{K}_{8,f}^\bullet$, and the zero-incidence cocycle gives its functoriality under composition. Naturality of Θ compares the two cubes without retyping it as exchange. This is a direct WDW-8 occupation of the registered RPI interface, not transport of Module C’s vacuum/full-Hilbert inhabitant through an asserted profile equality. $\square$

6.5 Calibration and symmetry scope

The compact hyperbolic core is not an approximately flat asymptotic target. No weak-field mass calibration and no Poincare/CPT limit is required for the core reconciliation theorem. The optional coordinates are typed as follows.

Coordinate	Classification	Scope reason
Weak-field positive-mass spin-zero calibration	Outside scope	The theorem reconstructs the curved compact hyperbolic Einstein–scalar history and makes no weak-field mass claim.
Local Lorentz recovery	Geometrically inherited	The reconstructed smooth Lorentzian metric and Levi–Civita connection provide local Lorentz frames; no quantum representation theorem is added.
Poincare recovery	Outside scope	The declared target is not flat or asymptotically flat.
CPT recovery	Outside scope	No local-QFT field algebra or flat-space particle sector is part of the core carrier.
History/record continuation	Established	The WDW-4A spacetime clock flow evaluated through WDW-3’s shifted-tower plaque equivalence, complete RPI role certificate, and Module C antecedents construct a nonidentity Moore category before the WDW-8 roles; differentiated evaluation derives the canonical seed formula, guarded-intersection uniqueness identifies the actual WDW-7 lineage, and the independently registered zero exchange ledger, full-state processes, typed occupation equalities, history-indexed identity cells, and strict four-cubes are proved in Theorem 6.13.

Proposition 6.14 (Scope-safe limit structure). *The out-of-scope calibration rows do not weaken any claimed core error coordinate. They prevent the compact curved theorem from being promoted to a weak-field particle, Poincare, CPT, or Standard-Model theorem.*

6.6 Integrated backreaction and continuation result

Theorem 6.15 (Integrated backreaction and continuation). *The scalar defect, full-envelope graph-attached Module-A response, history-dependent exact source family, separate reduced scalar action, direct homogeneous Einstein–source theorem, exact constrained family, explicit local covariant response, complete backreaction topology, Ward/exchange coordinate, and the RPI-occupied Module-C specialization with its upstream zero ledger, typed closed occupation, and history-indexed four-cubes are all constructed. The response is uniformly $O(\delta^2)$, its nonlinear remainder is $O(\delta^4)$, and every constraint, Ward, and total exchange residual is zero on the core recovery subtype.*

Proof. Lemmas 6.1–6.3 prove response, attachment, history-indexed source, and reduced-action population. Theorems 6.4 and 6.5 prove the exact family, complete tangent, explicit reconstruction, and estimate. Theorems 6.7 and 6.13 prove exact source lineage, exchange, and continuation. Proposition 6.14 prevents calibration overclaim. Together they establish the complete integrated result. $\square$

7. Integrated exhaustion, error closure, and endpoint theorem

7.1 First-divergence normal forms

Every attempted package in $\text{Cand}_g^{\text{fix}}$ is classified at its first standing-bearing divergence. The universe is fixed by Theorem 1.3 before the survivor is inspected.

Axis	Normal form	Classification
Parameter	Gauge/scale/clock	Eliminated: it does not change the quantum/-classical balance.
Parameter	Target-changing history/-source	Separate enrichment; not same-class recovery.
Parameter	Formal action symbol	Eliminated until an actual operator/state/domain family exists.
Parameter	Populated single action rate	Survives by Theorems 2.2 and 2.5.
Operator	Added primitive or multi-rate term	Enriched class; no same-class identity.
Phase	Inherited history jet called a wave phase	Eliminated by type.
Phase	Formal WKB exponential	Eliminated by nonpopulation.
Phase	Unregistered initial Hessian/higher jet	Inadmissible same-scope import: the handoff contains only the authenticated first phase jet.
Phase	Registered changed phase/error carrier	Standing-bearing scope enrichment, retained outside $\text{Cand}_g^{\text{fix}}$.
Phase	Generated affine authenticated first jet	Unique minimal prolongation by Theorem 1.4, then unique local solution by Lemma 3.1.
Amplitude	Blank/formal	Eliminated.
Amplitude	Unregistered nonconstant profile	Inadmissible same-scope import.
Amplitude	Registered profile carrier	Standing-bearing scope enrichment.
Amplitude	Generated unit source seed	Unique minimal prolongation by Theorem 1.4, then unique local solution by Lemma 3.3.
Reconstruction	Comparator-defined history	Eliminated as circular.
Reconstruction	Slicing-dependent history	Eliminated by Theorem 4.3.

Axis	Normal form	Classification
Reconstruction	Independent source-faithful history	Survives by Theorems 4.2–4.4.
Conditional carrier	Hidden spectator or BO factor	Eliminated: not generated.
Conditional carrier	WDW-7 line overwritten	Eliminated by carrier mismatch.
Conditional carrier	Quotient-descended actual WKB Cauchy-doublet line	Survives with $Y_{\delta,\tau}^{\text{rec}} = J_\tau \Psi_\delta$, the unique no-spectator conditioning map, defect-forced companion connection, zero separately typed endomorphism, and exact equation residual by Theorems 5.2–5.4.
Observable	Spatial-only or fixed-record conflation	Eliminated.
Observable	Scalar algebra plus typed complete record	Survives by Theorem 5.5.
Backreaction	Dropped by convention	Eliminated.
Backreaction	Zero-initial or off-shell response	Eliminated: violates the linearized constraint or the WDW-1 reconstruction domain.
Backreaction	Exact history-dependent source response	Survives with fixed-clock full-envelope metric-response population, Module-A graph attachment, a separate reduced scalar action, explicit homogeneous constraint inversion, exact scalar-source reevaluation, complete source tangent, direct FLRW Einstein–source equations, local reconstruction topology, and Ward/exchange by Lemmas 6.1–6.3 and Theorems 6.4–6.7.
Continuation	Downstream-solved or abstract-label seed	Eliminated: violates Module-C Proposition 33.5 or fails RPI occupation.
Continuation	Fixed-stage ODE for the derivative-losing ADM field	Eliminated: WDW-3 supplies no same-stage locally Lipschitz Hamiltonian vector field.
Continuation	Noncharacteristic transverse component or changed source/record coordinate	Changed continuation/response carrier or retained role coordinate; standing-bearing enrichment.
Continuation	WDW-4A spacetime scalar flow evaluated through the WDW-3 embedding-plaque equivalence	Survives by Lemma 6.9; differentiated evaluation derives $d\pi_{\text{pr}} X_{(\sqrt{h}/p_\phi, 0)}$ afterward, and the actual WDW-7 lineage and nonidentity Moore category follow.
Role processes	Undefined arrow, incomplete profile, or profile equality by declaration	Eliminated by target-category typing and independent fibre completeness.
Role processes	Eight constructed baseline/recovery functors and natural comparison	Survives by Lemma 6.11 and Theorem 6.13.
Exchange	Bare zero label inferred from later target readouts, with no upstream state ledger or history-incidence map	Eliminated: it does not inhabit Module-C Definition 22.2.
Exchange	Zero-valued object map with translated arrows before $J = 0$	Eliminated: the specified arrow misses its claimed endpoint whenever incidence is nonzero.
Exchange	Independently registered terminal zero ledger, typed target-occupation equalities, and history-indexed identity cells	Survives: the ledger and cocycle exist upstream, every target separately proves closed occupation, and each history generates a strict functorial sixteen-vertex cube.
Calibration	Curved result globalized to flat/SM claims	Eliminated by scope.

Axis	Normal form	Classification
Calibration	Explicitly out-of-scope rows	Survives by Proposition 6.14.

Theorem 7.1 (Complete fixed-domain normal form). *Every member of $\text{Cand}_8^{\text{fix}}$ reaches exactly one row in the preceding table at its first divergence. Every non-surviving same-scope row fails a fixed antecedent condition or the imported provenance rule. Every explicit enrichment changes a coordinate of the pre-fixed comparison class and is retained as a different target rather than being used to multiply the lawful same-scope residue.*

Proof. Theorem 1.3 first fixes the domain and proves that every standing-relevant variation is bookkeeping, failure, inadmissible import, or scope enrichment. Definition 1.5 then orders the WDW-specific coordinates: parameter role, operator lineage, phase/amplitude interface, reconstruction, conditional carrier, observable/record map, source response, spacetime clock flow, embedding evaluation, characteristic plaque, derived canonical seed, RPI role certificate and occupation, lineage comparator and orbit-identification test, continuation seed, terminal zero ledger and history incidence, role processes, target-occupation tests, history-indexed cells, four-cube, and scope. Two attempted packages have a first differing coordinate; the table instantiates all four imported dispositions at that coordinate. Theorem 1.4 constructs the source-generated phase/amplitude values before the table assigns their survival status. Hence neither exhaustiveness nor the survivor is inserted by definition. $\square$

7.2 Uniform constraint application

Theorem 7.2 (Uniform candidate testing). *Every candidate is tested with the same compact boundaryless target, parent source-family map and anchor C_{src} , WDW-2/3 quotient, WDW-4A profile clock and regular constraint chart, WDW-6 graph source and physical anchor, WDW-7 quotient coordinate, WDW-1 connection/reconstruction branch, Module-A Ward/source attachment discipline, exact history-dependent source rule, the WDW-4A spacetime scalar flow, WDW-3 shifted-tower embedding evaluation and full characteristic plaque, function-valued profile line, derived clock-normalized canonical seed, complete RPI role certificate, actual WDW-7 lineage and fixed Module-C history/record interface, the four role target types, terminal zero exchange monoid, upstream state ledger, role restrictions, zero history incidence, target-occupation equalities, lifted closed process, history-indexed cell and four-cube types, homogeneous recovery tube, explicit homogeneous right inverse and local reconstruction chart, common seminorms, and five-coordinate error register. No failed candidate is reclassified by changing one of those coordinates downstream.*

Proof. The coordinates are literal antecedents of Theorem 1.3 and Definition 1.5. The parameter, phase, amplitude, carrier, and response decisions occur only after they have been fixed. Every theorem above cites those same objects. Bivalence gives satisfaction or first failure at the same scope; a changed fixed coordinate is recorded openly as enrichment. $\square$

7.3 Error register

Error	Topology	Bound	Conclusion and uniformity
E_{HJ}	Pulled-back $p_{K,r}$	$C_{\text{HJ}}\delta^2$	Vanishing on K , the clock slab, and the C^r lapse unit ball.
E_{Ein}	Quotient-aligned C^{s-2} and C^{s-3} norms	0	Exact on the common reconstructed slab.
E_{mat}	Conditional residual on the descended quotient line	0	Exact on the recovery line over $I \times K$.
E_{obs}	Scalar-algebra/record probe metric	0	Exact on the declared finite probe class.
E_{back}	C^{k-1} canonical response, explicit reconstruction norm, complete source semi-norm, nonlinear and exchange residuals	$C_{\text{back}}\delta^2$	Vanishing on $\mathcal{N}_{\text{hom}}$, $I \times K$, and the exact history-indexed source family.

The constants depend on the fixed chart, compact tube, lapse regularity, source coefficients, finite normalization profile, amplitude derivatives, and physical constants. They do not depend on δ .

Theorem 7.3 (Complete error closure). *All five analytic coordinates are defined and bounded, and they vanish on the declared recovery direction. In the notation of the inherited interface,*

$$\sigma_{\text{est}} = \textbf{ESTIMATED-WDW8}.$$

No out-of-scope calibration row is counted as a sixth error or used to discharge an analytic coordinate.

7.4 Singleton lawful survivor at the declared scope

Theorem 7.4 (Fixed-domain lawful recovery residue). *For the complete fixed candidate universe,*

$$\left| \text{Real}_8^{\text{law}} / \sim_8 \right| = 1.$$

The survivor is the constraint-scaled, homogeneous-cylinder, generated-affine-first-jet, generated-unit-amplitude, cutoff-supported, quotient-descended, constraint-tangent, source-generated package constructed above.

Proof. Population is Theorem 2.5 and Lemma 3.4. Any positive single rate is equivalent to δ by Theorem 2.1; the operator is then fixed by (2). The phase-jet theorem makes the inherited first phase jet sufficient for the target history, and Theorem 1.4 constructs its unique no-higher-jet and no-profile prolongations. A different unregistered higher jet/profile is an import; a registered one changes the phase/error carrier and is an enrichment, by Theorem 1.3. The HJ and transport Cauchy problems then have unique solutions. Hamilton characteristics and WDW-1 reconstruction are unique before the failure boundary, modulo exactly the WDW-2/3 presentation arrows.

Theorem 5.2 makes the recovery conditioning map unique from I_6 , the actual trace $J_\tau \Psi_\delta$, and no-spectator generation; the exact inhomogeneous companion identity then fixes the connection response and zero endomorphism. The represented algebra is $\mathbb{C}I$ and the record is fixed by the reconstructed history. Equation (51) fixes the profile-clock-transverse homogeneous constraint correction. Adding a kernel component either is a WDW-2/3 constraint direction and hence equivalent, or changes the physical Cauchy/source target. Lemmas 6.1 and 6.2 fix the Module-A response base and attachment before the triplet is represented. Equations (62) and (63) then fix the changed scalar fibre and the full source

tangent; any extra source term is probe-null skin or changes the source family. Lemma 6.9 constructs the lawful RPI occupation, discharges the complete Module-C antecedent certificate, identifies the anchor flow with WDW-7’s actual occupied lineage, and fixes the positively populated clock-normalized characteristic seed because $Np_\phi/\sqrt{h} = 1$ forces $N = \sqrt{h}/p_\phi$ while proper-spatial shift vanishes in the quotient; Lemma 6.11 fixes the four target profiles, and Theorem 6.13 fixes their full-state-factored process and naturality coordinates, verifies their occupation of the independently registered closed ledger, and proves the history-indexed exchange cubes strict. The forced response equation, explicit local reconstruction and constructed Module-C specialization are therefore unique modulo the same lawful arrows. All errors and first-failure data are fixed. No second inequivalent lawful same-scope package remains. $\square$

Remark 7.5 (Exact uniqueness strength). The theorem eliminates every false same-scope candidate by the complete kernel-faithful constraint intersection. It does not deny that an openly changed source, initial jet, amplitude carrier, multi-rate operator grammar, matter factor, observable algebra, classical history, RPI occupation, continuation seed, target category, exchange carrier, or recovery seminorm can define a different target. Such enrichments require their own population and error theorem and do not compete in $\text{Real}_g^{\text{law}}$.

7.5 Seventeen-node theorem chain

Node	Theorem burden	Closing result
W8.1	Inherited data and fixed scope	Theorem 1.1
W8.2	Parameter exhaustion	Theorem 2.1
W8.3	Positive family population	Theorems 2.2, 2.5
W8.4	Common domains/topologies	Theorems 2.2, 2.5
W8.5	Phase–amplitude population	Theorem 1.4; Lemmas 3.1, 3.3, and 3.4
W8.6	Exact HJ identity/estimate	Theorems 3.5, 3.6
W8.7	Canonical-to-covariant constructor	Theorem 4.2
W8.8	Redescription/refoliation independence	Theorem 4.3
W8.9	Einstein/source recovery	Theorem 4.4
W8.10	Connection/curvature recovery	Theorem 4.5
W8.11	Profile-clock conditional recovery	Lemma 5.1; equation (19); Theorems 5.2–5.4
W8.12	Observable/record recovery	Theorem 5.5
W8.13	Backreaction/source exchange	Lemmas 6.1–6.3; equation (51); Theorems 6.4–6.7
W8.14	Characteristic-generated RPI Module-C specialization, actual-lineage projection, calibrated exchange cube, and limit scope	Definition 6.8; Lemmas 6.9, 6.11; Theorem 6.13; Proposition 6.14
W8.15	Complete package exhaustion	Theorem 1.3; Definition 1.5; Theorems 7.1, 7.2
W8.16	Covariant–canonical reconciliation	Theorems 7.3, 7.4
W8.17	G8 and WDW-9 export	Theorem 7.6 below

7.6 Endpoint and subsequent handoff

Define

$$G_8 := (\text{Scope}_8, \text{Imports}_8, \text{Pop}_8, \text{HJ}_8, \text{Cov}_8, \text{Mat}_8, \text{ObsRec}_8, \\ \text{Back}_8, \text{Limit}_8, \text{Residue}_8, \text{Boundary}_8, \text{Cert}_8).$$

Each coordinate is defined by the results collected in the preceding theorem chain.

Theorem 7.6 (Endpoint theorem and WDW-9 handoff). *All seventeen conditions in the dependency matrix are satisfied at the exact fixed scope of Theorem 1.3 and Definition 1.5. The lawful live recovery residue is nonempty and singleton; all five claimed analytic errors vanish; weak-field, Poincaré, and CPT coordinates lie outside the declared scope. Therefore*

$$G_8 = \textbf{PASS}, \quad \text{Authorize(WDW-9)} = \text{TRUE}$$

with the lawful recovery subtype, all failure data, and all nonclaims retained as standing-bearing export coordinates.

Proof. Population is Theorem 2.5; error closure is Theorem 7.3; residue classification is Theorem 7.4; and scope-safe calibration is Proposition 6.14. These are precisely the fixed handoff antecedents. No downstream WDW-9 conclusion is used. $\square$

Corollary 7.7 (Principal theorem). *Theorem 0.1 holds.*

8. Dependency matrix and robustness analysis

8.1 Theorem dependency matrix

Node	Requirement	Conclusion	Evidence
G8-01	Exact $I_{6 \rightarrow 8}$ import	Established	Theorem 1.1
G8-02	Candidate universe fixed	Established	Theorem 1.3; Definition 1.5; Theorem 7.1
G8-03	Positive non-anchor population	Established	Theorems 2.2, 2.5
G8-04	Common domains/topologies	Established	Theorem 2.2; equation (4)
G8-05	Phase–amplitude population	Established	Theorem 1.4; Lemmas 3.1, 3.3, and 3.4
G8-06	HJ residual bound	Established	Theorems 3.5, 3.6
G8-07	Independent reconstruction	Established	Theorem 4.2
G8-08	Slicing/refoliation independence	Established	Theorem 4.3
G8-09	Einstein/source recovery	Established	Theorem 4.4
G8-10	Fixed connection recovery	Established	Theorem 4.5
G8-11	Matter conditioning/evolution	Established	Lemma 5.1; Theorems 5.2, 5.3, and 5.4
G8-12	Observable/record comparison	Established	Theorem 5.5
G8-13	Backreaction/source exchange	Established	Lemmas 6.1–6.3; Theorems 6.4, 6.5, and 6.7
G8-14	Module-C specialization and calibration scope	Established	Definition 6.8; Lemmas 6.9, 6.11; Theorem 6.13; Proposition 6.14
G8-15	Uniform candidate testing	Established	Theorems 1.3, 7.1, and 7.2
G8-16	Live residue computed	Established	Theorem 7.4
G8-17	G8 endpoint	Established	Theorem 7.6

8.2 Potential objections

Each entry states an objection in its strongest form and resolves it by reference to the relevant construction. Labels such as “semiclassical”, “constraint-derived”, or “kernel-faithful” are not used as substitutes for the cited results.

Objection and resolution 8.1 (A. The singleton seed is relabeled as a family). The family has actual non-anchor operators (2) and actual non-anchor source elements (11) for every $0 < \delta \leq \delta_0$. The physical

seed remains at the isolated parameter 1. Theorem 2.5 proves source attachment without asserting state convergence.

Objection and resolution 8.2 (B. The renormalization address is smallness). The parameter is δ , not μ . Theorem 2.1 uses WDW-6 scale equivalence to eliminate μ before the operator family is defined.

Objection and resolution 8.3 (C. The clock or ϵ_* is smallness). The profile clock and ϵ_* are fixed in every package. Varying τ is evolution; varying ϵ_* changes the source history. Neither occurs in (2).

Objection and resolution 8.4 (D. The inherited phase jet is called the phase of Ψ). The phase S is the unique solution of the Hamilton–Jacobi Cauchy problem in Lemma 3.1. Proposition 3.7 proves its initial-jet relation afterward. No equality is a definition.

Objection and resolution 8.5 (E. A formal WKB ansatz substitutes for population). Equation (11) uses one fixed compactly supported smooth cutoff and is therefore literally in the domain of E_{hom} for every positive δ . Lemma 3.4 proves that the cutoff is identically one on a larger tube around every estimated point, so the exact operator identity acts on an actual WDW-5/6 source element without acquiring a local cutoff commutator.

Objection and resolution 8.6 (F. The covariant history is assumed in reconstruction). Theorem 4.2 constructs γ_S^{cov} from S and canonical Hamilton equations. Only Theorem 4.4 applies the imported history as comparator.

Objection and resolution 8.7 (G. Constraints are mistaken for full Einstein evolution). Lemma 4.1 includes the Hamilton evolution equations. WDW-1’s ten-component theorem then reconstructs the full Einstein–Klein–Gordon equations in Theorem 4.2. Constraint vanishing alone is not used.

Objection and resolution 8.8 (H. No common topology makes errors vanish). Equation (4) gives the HJ topology. For backreaction, equations (34), (47), (68), (71), and (77) separately define the lapse class, complete source seminorm, history chart, covariant-response norm, and uniform error. No undefined solution-quotient norm or derivative is used. The complete register appears in Theorem 7.3.

Objection and resolution 8.9 (I. Recovery depends on lapse, shift, or slicing). Theorem 4.3 applies the accepted WDW-2/3 groupoid and characteristic-refoliation theorems. Lapse and shift are presentation coordinates; the error seminorm is uniform over a bounded lapse class.

Objection and resolution 8.10 (J. A new connection is silently used). Theorem 4.5 recovers the fixed minimal metric branch and its unique Levi–Civita derivative. No independent connection variable appears.

Objection and resolution 8.11 (K. The conditional equation is tautological or a matter Hamiltonian is imported). Theorem 2.5 first proves that no Born–Oppenheimer factor carrier or polarization is generated. Equation (19) applies the actual WDW-7 trace to the populated source element and retains both value and normal response; no point evaluation is used as a state. Theorem 5.2 then defines the recovery conditioning map on the WDW-6 quotient through I_6 and proves its no-spectator uniqueness. Reapplication of the scaled constraint gives the inhomogeneous companion identity (25) and independently fixes its forcing before ∇^{rec} is defined. Theorem 5.3 proves coefficient transport and physical-line intertwining. The separately typed zero endomorphism is then forced on the generated line. The residual (30) is the actual left-minus-right equation residual and is identically zero. The quantum-potential defect remains explicit in the companion forcing, E_{HJ} , and Theorem 6.15; it is not split into cancelling matter terms.

Objection and resolution 8.12 (L. WDW-7’s zero Hamiltonian is overwritten). Section 5 types $\mathcal{L}_\tau^{\text{C}}$ and $\mathcal{L}_{\delta,\tau}^{\text{rec}}$ separately. The physical WDW-7 endomorphism remains exactly zero. Theorem 5.3 proves exact intertwining of the separately typed recovery zero endomorphism and explicitly denies any promotion to a nonzero full-Cauchy Hamiltonian.

Objection and resolution 8.13 (M. Spatial descent substitutes for Hamiltonian-constraint descent). The complete record is evaluated at the WDW-4A function-valued profile-clock intersection and is

preserved by the WDW-3 full characteristic quotient. Theorem 5.5 uses both spatial and refoliation descent.

Objection and resolution 8.14 (N. Backreaction is off shell, Ward compatibility is formal, or the HJ remainder is counted twice). The metric readout (38) is defined on every compact metric protocol through the fixed-clock ADM projection (35); shift, shear, and mean-zero modes are recorded in its kernel. The scalar clock is held fixed throughout that Module-A variation, so no δN term is omitted. The response class precedes the complete metric-response relation and graph attachment. Only afterward is the separate reduced action (45) introduced; its scalar variation holds the multiplier N fixed and produces exactly $-N\partial_\phi r_\delta$. Equations (51)–(56) construct an exact carrier-preserving constraint family. Equations (58)–(61) then prove every homogeneous Einstein, scalar and source column directly; no new full WDW-1 pointed strong lift or Dirac ledger is claimed. Crucially, equation (62) reevaluates the scalar source on each changed history, and (63) retains $D\text{Src}_\phi^{\text{hom}}\zeta + C_\delta^Q$ rather than only the quantum term. The covariant response is the derivative of the explicit local map (69); WDW-1 is neither extended to the deformed family nor differentiated. Definition 6.8 uses WDW-4A’s spacetime scalar flow and WDW-3’s shifted-tower evaluation map to construct the full plaque, occupied clock line, prefixes, complete RPI role certificate, and normalized seed. Differentiated evaluation derives $N_\phi = \sqrt{\hbar}/p_\phi$ and the canonical characteristic formula only after the path exists. Lemma 6.9 proves the Module-C antecedents and only then identifies its anchor flow with WDW-7’s actual occupied lineage and prefix. Only its Moore category is used upstream; it contains no WDW-8 parameter, F_8 , solved metric, or Hilbert flow. Its terminal zero monoid, state ledger, role restrictions, and history incidence are independently registered in (87)–(98) before target evaluation. Equations (111)–(138) then construct the endpoint objects and all eight baseline/recovery process arrows, their full-state factorization, composition laws, and four naturality squares. Equation (127) then checks Ward, Hilbert, record, and compatibility closure without subtracting quotient objects or changing the upstream ledger. The separately named closed/enriched lifts, predicted and actual endpoints of each history, identity cells, and functorial sixteen-vertex cubes are constructed in (128)–(137). No vacuum/nonvacuum or unrelated full-Hilbert/rank-one profile equality is asserted. The parameter ϵ is explicitly excluded from upstream incidence. Theorem 6.5 then bounds the complete canonical, covariant, and source response by $O(\delta^2)$ and the nonlinear remainder by $O(\delta^4)$. E_{HJ} measures the scalar defect; E_{mat} uses its exact Cauchy forcing; and E_{back} measures the attached constrained response and exact source/exchange residuals.

Objection and resolution 8.15 (O. A weak-field calibration is generalized). No weak-field calibration is claimed. It lies outside the compact curved target, as recorded in Proposition 6.14.

Objection and resolution 8.16 (P. Local Lorentz/Poincaré/CPT recovery is globalized). Local Lorentz geometry is inherited from the reconstructed Lorentzian metric. Poincaré and CPT recovery lie outside the declared scope because no flat local-QFT target is present.

Objection and resolution 8.17 (Q. Errors are small only at the anchor or one clock instant). Every bound is uniform on the compact recovery tube K , compact clock interval I , spacetime slab $\mathcal{K}$, and specified lapse unit ball. The physical anchor is not used as the limiting point.

Objection and resolution 8.18 (R. Asymptotic notation has no remainder theorem). The exact identity (13) produces the remainder. Theorems 3.6, 5.4, and 6.5 give explicit powers, constants, topologies, and ranges.

Objection and resolution 8.19 (S. The lawful universe was replaced by a definitionally minimal class). Theorem 1.3 imports the kernel, fixed-domain exhaustion, same-scope operator exhaustion, and bivalence before Definition 1.5. That definition forms the broad attempted class $\text{Cand}_g^{\text{fix}}$, including formal and failed routes; it defines $\text{Real}_g^{\text{law}}$ only by later survival of the fixed conditions. Its coordinates expressly include the WDW-4A spacetime-flow type, WDW-3 shifted-tower evaluation and plaque types, evaluated clock line, derived canonical-seed test, RPI role-certificate and occupation tests, inherited WDW-7 lineage comparator, terminal zero exchange ledger and history-incidence types, four target-category and process types, typed closed-occupation tests, history-indexed cell and cube types, and target-failure tests. It does not prescribe which candidate satisfies the physical occupation tests. The phase-jet theorem

authenticates the complete first phase jet, and Theorem 1.4 constructs the no-spectator prolongations before the normal-form table is applied. An unregistered higher jet/profile is an inadmissible same-scope import; a registered carrier, probe, seed, or continuation change is an explicit enrichment. Theorem 7.1 has separate continuation, process, and exchange axes and Theorem 7.4 computes the populated lawful quotient.

Objection and resolution 8.20 (T. A conditional theorem is promoted as positive recovery). Theorem 2.5 proves actual population; Theorem 7.4 proves a nonempty live quotient; and Theorem 7.3 closes every claimed analytic coordinate before the endpoint conclusion.

Objection and resolution 8.21 (U. WDW-8 alters an upstream result). The family begins from the accepted WDW-6/7 object and introduces a separate recovery comparison germ. It changes no upstream domain, anomaly, form, state, observable, or clock theorem.

Objection and resolution 8.22 (V. Kernel terminology replaces population). Theorem 1.3 fixes the candidate domain but supplies no occupant. Positive population is independently proved by Theorems 2.2, 2.5, and equation (11); source response is independently populated by Lemmas 6.1–6.3 and Theorem 6.4. Kernel constraints classify those actual objects and their rivals; they never substitute for existence.

Objection and resolution 8.23 (W. An empty residue is hidden behind uniqueness). Theorem 2.5 exhibits the survivor before Theorem 7.4 computes its quotient. The cardinality is one, not an implication over an empty class.

Objection and resolution 8.24 (X. The open nonzero full-Cauchy Hamiltonian is made load-bearing). It does not appear in the theorem chain. The accepted source-generated WDW-7 line retains its exact zero covariant endomorphism. The separately typed recovery line carries the quotient-descended zero endomorphism of (29); it is never promoted to a full-Cauchy operator. The full-Cauchy extension remains open.

Objection and resolution 8.25 (Y. RPI standing is assigned, downstream data backflow into the seed, or the exchange cube is tautological). Equations (79)–(83) construct the seed from the accepted WDW-4A spacetime flow and WDW-3 evaluated plaque. The scalar-response identity (100) proves normalization, nonzero seed, and tangency; it does not assign them as role labels. The twelve coordinates in (89) have literal WDW-4/6/7 occupants, and Lemma 6.9 proves $K_D \wedge \text{FixedAdmRealPers}_D \wedge \text{RoleSeq}_D$ before invoking RPI or Module C. The upstream packet contains F_7 and only the independently declared zero exchange ledger, but no F_8 or downstream physical readout. The compatibility objects K_u^0, K_u^δ are separately defined in (101). The terminal ledger is therefore not inferred from those solved targets. Equation (127) instead tests whether each independently completed process inhabits the fixed closed coordinate; any failure rejects the candidate and cannot rewrite its ledger. Equations (132)–(137) then build the strict history-indexed cells and cubes. Finally, Definition 1.5 and Theorem 7.1 include the continuation and exchange axes in the fixed candidate universe. None of these conclusions follows by label assignment.

Objection and resolution 8.26 (Z. An arbitrary analytic curve is called the WDW-7 lineage, targets are synchronized only by clock, or Hilbert incidence is ill typed). No analytic-mean flow-box curve occurs in the construction. The seed is the evaluated WDW-4A spacetime embedding flow (81), formed before any WDW-8 solved target. WDW-3 maps that actual path into the characteristic plaque and only then derives $d\pi_{\text{pr}}X_{(\sqrt{h}/p_\phi, 0)}$ by (86). The evaluated path has $\phi(\cdot) = u\mathbf{1}$; WDW-4A's unique guarded-intersection theorem therefore proves (95), identifying it with WDW-7's actual s_u and ρ_u only after the seed exists. The object maps in (115) take the complete state $x = (z, C_{\text{src}}, I_6, F_A^{(0)}, \tau, r, \dots)$ as input, and $\text{Dev}^\bullet(f)$ is the induced process of the entire upstream segment; a common clock label alone cannot authorize a target. Finally, $\mathcal{H}_u^\bullet$ in (126) applies the connection and generator to the constructed transported nonzero section $Y_u^\bullet$, never to the set $\mathbb{P}(\mathcal{L}_u^\bullet)$. Parallel transport and the separately typed zero generator prove the two components zero after the readout is defined.

Objection and resolution 8.27 (AA. Smooth-tame derivative loss is promoted to a Banach ODE, or

the exchange pair misses its endpoint). No smooth-tame-to-Banach-flow implication occurs. WDW-4A first supplies the ordinary finite spacetime flow F_t^ϕ ; WDW-3 then evaluates its Cauchy embedding path using the exact shifted Sobolev tower and Nash–Moser equivalence. In the resulting clock-line chart the seed is literally ∂_u , with zero Lipschitz constant, and its translation is the displayed map (93). The derivative-losing canonical field appears only in the differentiated identity (86).

Independently, the exchange object is the explicitly declared closed object $\mathbf{0}_D$, not a zero-valued object map paired with nonzero translated arrows. The lifted predictor and enriched actual process both preserve the unique role ledger, so (132) has compatible source and target before any physical closure theorem. The target tests then establish lawful occupation; no incompatible pair is altered after the fact.

Objection and resolution 8.28 (AB. The upstream ledger is absent, quotient-valued defects are subtracted, or the cube has the wrong source). The raw packet carries the literal zero map $\ell_{D,8}^{\text{pl}}$ in (88), and its unique descent is the full-interior Module-C ledger $\ell_{D,8}^{\text{cl}} : I_{D^{\text{pl}}} \rightarrow \mathbf{0}_D$ in (91). After the Moore category is formed, $J_{D,8}^{\text{cl}}(f) = 0_D$ and (98)–(99) discharge Module-C Definition 22.2 before any target evaluation. No record or compatibility quotient elements are subtracted. Their physical closure is the typed endpoint-preservation equality in (127). Finally, $E_8^{X,\bullet}(f)$ is indexed by a history and compares the lifted closed prediction from its actual source with its actual endpoint, exactly as in Module-C Definition 22.5. The cube $\mathcal{K}_{8,f}^\bullet$ is consequently history-indexed and respects composition; it is not an objectwise anchor cube.

8.3 Reproducibility boundary

The publication check confirms theorem and dependency order, objection dispositions, error coordinates, status transitions, first-failure rows, non-anchor population, Cauchy-doublet conditioning, full-envelope projection and graph attachment, history-dependent source assignment, local reconstruction, homogeneous constraint inversion, canonical-seed exhaustion, process laws, recovery naturality, typed physical occupation, history-indexed cells, strict four-cubes, nonclaims, and the WDW-9 handoff. These finite checks verify internal coherence; they do not replace the mathematical proofs.

9. Exact boundary of the result

Theorem 7.6 is a positive recovery theorem, but it is not a universal semiclassical-gravity theorem. This section states its exact domain directly.

9.1 The physical anchor and the recovery germ are not a norm-convergent state path

The parameter carrier is the disjoint union

$$\mathcal{P}_8 = \{1\}_{\text{phys}} \sqcup (0, \delta_0]_{\text{rec}}.$$

The point 1_{phys} retains the accepted WDW-6 operator and authenticated physical state. The recovery component carries the constraint-scaled operators and the phase–amplitude elements used in the estimates. Source attachment means equality of the accepted operator at the physical anchor, common source lineage, the same pointed classical jet, the same clock, the same classical principal symbol, and the same normalization character. It does *not* mean

$$\Psi_\delta \longrightarrow F_A^{(0)}$$

in a Hilbert, distributional, source, or graph topology. No such convergence is proved or used. A recovery theory requiring a continuous physical-state interpolation is a strictly enriched WDW-8 class and remains open.

This separation is forced by the inherited rank-one physical quotient: multiplying its unique ray by a scalar would not populate a genuine approximation family. The recovery component supplies comparison elements in the actual common source domain; it does not manufacture extra physical states.

9.2 Homogeneous compression is the declared target, not a full-superspace estimate

Every analytic residual in this paper is evaluated after the source-generated homogeneous-cylinder compression

$$R_{\text{hom}} \widehat{\mathcal{H}}_{\delta}^{\text{ren}} E_{\text{hom}}.$$

Uniformity is over a compact neighborhood $K \Subset U_{\text{hom}}$, the specified bounded lapse class, and $0 < \delta \leq \delta_0$. No estimate is asserted in transverse inhomogeneous superspace directions. Those directions remain present in the parent WDW-6 operator and WDW-7 Cauchy germ; they are not estimated by the WDW-8 comparison seminorm.

Accordingly, the result recovers the accepted compact hyperbolic homogeneous Einstein–Klein–Gordon history from its canonical data. It does not classify arbitrary geometries, arbitrary WDW-6 source elements, or arbitrary perturbations of the homogeneous cylinder.

9.3 Exact equality and asymptotic control are kept distinct

The five error coordinates have different logical sources:

Coordinate	Certified value	Reason
E_{HJ}	$O(\delta^2)$	Exact WKB substitution after HJ and transport cancellation.
E_{Ein}	0	Canonical characteristic reconstruction and Cauchy uniqueness.
E_{mat}	0	Actual WDW-7 value/normal-response trace, defect-forced companion connection, separately typed zero endomorphism, and genuine equation residual.
E_{obs}	0	Same represented scalar algebra and same complete profile-clock volume record.
E_{back}	$O(\delta^2)$	Fixed-clock full-envelope metric attachment, separate reduced scalar action, exact history-dependent scalar/quantum source, direct homogeneous Einstein–source equations, explicit correction and local reconstruction, complete source seminorm, exact Ward/exchange residual, and $O(\delta^4)$ nonlinear remainder.

The zero entries are not limits inferred from the asymptotic entries. They follow from quotient descent, constraint-generated parallel transport, source ancestry, and independent reconstruction. Conversely, the nonzero-order entries are not promoted to exact identities.

9.4 What the theorem does not claim

The following are explicit nonclaims:

- N1.** no global clock, global Hamilton–Jacobi branch, caustic crossing, or singular-boundary continuation;
- N2.** no uniqueness across openly changed sources, phase/amplitude carriers, multi-rate operator grammars, matter factors, comparison probes, or classical targets;
- N3.** no positive-frequency projection and no nonzero full-Cauchy Hamiltonian;
- N4.** no full-superspace, ultraviolet, continuum-removal, or renormalization-group limit;
- N5.** no general locally covariant, state-dependent, renormalized quantum stress tensor beyond the specified homogeneous order- δ^2 Module-A response, and no new full field-theoretic WDW-1 strong lift for the deformed family;
- N6.** no weak-field, asymptotically flat, Poincaré, CPT, Standard-Model, or scattering recovery theorem;
- N7.** no WDW-9 semiclassical solution-selection, decoherence, arrow-of-time, or phenomenology result.

9.5 Failure surfaces and fail-closed continuation

The certified component ends at the first occurrence of any of the following:

$$F_8 = F_{\text{source}} \cup F_{\text{domain}} \cup F_{\text{phase}} \cup F_{\text{amp}} \cup F_{\text{rec}} \cup F_{\text{clock}} \cup F_{\text{srcresp}} \cup F_{\text{cont}} \cup F_{\text{uniform}}.$$

These denote, respectively, loss of source attachment, common-domain failure, a characteristic or caustic phase surface, loss of amplitude positivity or regularity, failure of canonical reconstruction, failure of the profile-clock guard, loss of the homogeneous right-inverse or local reconstruction chart, history-indexed source map, Module-A attachment/Ward chart, loss of the WDW-2/3 regular constraint chart, nonzero scalar momentum, function-valued profile slice, characteristic seed, RPI occupation, actual-lineage identification, target-fibre completeness, process functoriality, incidence cocycle, or exchange-cube typing, or loss of compact uniform bounds. At F_8 the corresponding error coordinate is no longer estimated. The theorem makes no continuation claim through that surface.

Theorem 9.1 (Boundary-complete interpretation). *Within the exact fixed domain of Theorem 1.3 and Definition 1.5, and only before F_8 , the endpoint theorem proves a populated, source-attached, covariant–canonical recovery germ with the error register of Theorem 7.3. The isolated-anchor, homogeneous-subtype, local-clock, no-spectator, exact-source-fibre, constraint-tangent, full-protocol-envelope, and Ward-compatible qualifications are theorem hypotheses, not removable prose. Deleting one changes the comparison class.*

Scope summary. The endpoint theorem holds for the fixed Branch-A, profile-clock, compact-hyperbolic homogeneous, source-attached, single-rate no-spectator lawful recovery class. All changed-target, global, transverse-superspace, singular, ultraviolet, asymptotically flat, nonzero-full-Cauchy-Hamiltonian, and WDW-9 selection claims remain open.

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Source note

References to WDW-1–7 and the AASC support chain identify the declared source lineage. The mathematical content is contained in the proofs above.